

Exam 7 High-Level Summaries

Fall 2026 Sitting

Rising Fellow



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Mack - Benktander

Credible Claims Reserves: The Benktander Method

Key Ideas

This paper is all about the Benktander method, with the following key takeaways:

- It is the Bornhuetter-Ferguson (BF) method run twice.
- Its reserve estimate can be viewed as a credibility-weighting of the chain ladder (CL) and BF
- The credibility weight is %Paid ($Z = p = 1 - q$), and it often outperforms the BF and CL

Benktander Method

The Benktander method can be calculated as a second iteration of the BF procedure or as a credibility-weighting of the Chain Ladder and Expected Loss ultimates (see the “Mack-Benktander – Benktander Method” recipe).

Benktander as a second iteration of the BF procedure

Iteration 1 – Bornhuetter-Ferguson:

$$U_{BF} = C_k + q_k U_0$$

$$Ult_{BF} = Loss + (1 - \%Paid) \times Prem \times ELR$$

Iteration 2 – Benktander:

$$U_{GB} = C_k + q_k U_{BF}$$

$$U_{GB} = Loss + (1 - \%Paid) \times Ult_{BF}$$

Benktander as a credibility-weighting of the Chain Ladder and Expected Loss Ultimates

Chain Ladder Ultimate:

$$U_{CL} = \frac{C_k}{p_k}$$

$$Ult_{CL} = Loss \times CDF$$

Benktander:

$$q_k = 1 - \frac{1}{CDF}$$

$$U_{GB} = (1 - q_k^2) U_{CL} + q_k^2 \times U_0$$

$$Ult_{GB} = [1 - \%Unpaid^2] \times Ult_{CL} + \%Unpaid^2 \times Prem \times ELR$$

Benktander as a credibility-weighting of the Chain Ladder and BF Reserves

$$R_{GB} = (1 - q_k) R_{CL} + q_k \times R_{BF}$$

$$Resv_{GB} = [1 - \%Unpaid] \times Resv_{CL} + \%Unpaid \times Resv_{BF}$$

Advantages of the Benktander Method

- Outperforms the BF and Chain Ladder methods in many circumstances with a smaller MSE
- The MSE of the Benktander reserve is almost as small as that of the optimal credibility reserve

Iterated BF Method

The Benktander method is a second iteration of the BF procedure. This is how the iteration works:

1. Start with an ultimate loss estimate, $U^{(m)}$. For $U^{(0)}$, use the expected loss estimate.
2. Apply the BF procedure to get a new loss reserve estimate:

$$\boxed{R^{(m)} = q_k U^{(m)}} \quad Resv^{(m)} = \%Unpaid \times Ult^{(m)}$$

3. Get a new ultimate loss estimate by adding the losses-to-date to the reserve. This is the starting ultimate for the next iteration:

$$\boxed{U^{(m+1)} = C_k + R^{(m)}} \quad Ult^{(m+1)} = Loss_k + Resv^{(m)}$$

The ultimate loss estimate ($U^{(m)}$) can be rearranged as a credibility weighting of the Chain Ladder ultimate (U_{CL}) and expected loss ultimate (U_0). Also, the loss reserve estimate ($R^{(m)}$) can be rearranged as a credibility weighting of the Chain Ladder reserve (R_{CL}) and the BF reserve (R_{BF}):

$$\boxed{U^{(m)} = (1 - q_k^m) U_{CL} + q_k^m \times U_0} \quad \boxed{R^{(m)} = (1 - q_k^m) R_{CL} + q_k^m \times R_{BF}}$$

m	Starting Ultimate ($U^{(m)}$)	New Reserve ($R^{(m)}$)
0	$U_0 = Prem \times ELR$	
		$\xrightarrow{\times \%Unpaid}$
		$R_{BF} = q_k U_0$
	$U_{BF} = Loss + R_{BF}$	$\xleftarrow{+ Loss}$
1	$U^{(1)} = (1 - q_k^1) \times U_{CL} + q_k^1 \times U_0$	
		$R_{GB} = q_k U_{BF}$
		$R^{(1)} = (1 - q_k^1) R_{CL} + q_k^1 \times R_{BF}$
		Benktander
2	$U_{GB} = Loss + R_{GB}$	
	$U^{(2)} = (1 - q_k^2) \times U_{CL} + q_k^2 \times U_0$	
		$R^{(2)} = q_k U^{(2)}$
		$R^{(2)} = (1 - q_k^2) R_{CL} + q_k^2 \times R_{BF}$
$\vdots$	$\vdots$	$\vdots$
∞	$U^{(\infty)} = U_{CL}$	$R^{(\infty)} = R_{CL}$

As the number of iterations increases, the weight on the chain ladder method increases until it converges to the chain ladder method entirely (as $m \rightarrow \infty$).

Recipes for Calculation Problems

- Benktander Method

Overview

The reserve estimate method in Hürlimann is a credibility-weighted method that's very similar to the Mack (2000) method. The key difference is that Hürlimann uses expected incremental loss ratios (m_k) to specify the payment pattern instead of using LDFs calculated directly from the losses.

Hürlimann uses two new reserving methods based on the loss ratio payout factors, p_i :

- Individual Loss Ratio Reserve (R^{ind}) – Similar to the chain ladder method
- Collective Loss Ratio Reserve (R^{coll}) – Similar to the Bornhuetter-Ferguson (BF) method

The key idea from Hürlimann is that R^{ind} and R^{coll} represent extremes of credibility on the actual loss experience and we can calculate a credibility-weighted estimate that minimizes the MSE of the reserve estimate.

Credible Loss Ratio Claims Reserve

Individual Loss Ratio Claims Reserve (R^{ind})

$$R^{ind} = \frac{q_i \cdot Loss_i}{p_i}$$

$$R^{ind} = \frac{\%Unpaid_{AY} \cdot Loss_{AY}}{\%Paid_{AY}}$$

R^{ind} – 100% credibility on losses-to-date

Collective Loss Ratio Claims Reserve (R^{coll})

$$R^{coll} = q_i \cdot Prem \cdot ELR$$

$$R^{coll} = \%Unpaid_{AY} \cdot Premium_{AY} \cdot ELR$$

R^{coll} – 0% credibility on losses-to-date

Credibility-Weighted Reserve Estimate

We can calculate a new, credibility-weighted estimate, based on R^{ind} and R^{coll} , that minimizes the mean squared error (MSE) and variance of the loss reserve estimate.

$$R_i = Z_i \cdot R_i^{ind} + (1 - Z_i) \cdot R_i^{coll}$$

Hürlimann uses three different credibility methods:

- Benktander (also in Mack (2000))
- Neuhaus
- Optimal credibility weighting (minimizes MSE)

Method	Z_i
Benktander(Z_i^{GB})	$Z_i^{GB} = p_i$
Neuhaus(Z_i^{WN})	$Z_i^{WN} = p_i \cdot ELR$
Optimal(Z_i^{opt})	$Z_i^{opt} = \frac{p_i}{p_i + \sqrt{p_i}}$

Advantages over the Mack (2000) approach

- Straightforward calculation of the optimal credibility weight
- Different actuaries get the same result using the collective loss ratio claims reserve with the same premiums (BF method requires an ELR assumption)

Advantage of the optimal credibility weight reserve (R^{opt})

- Minimizes MSE and variance of the loss reserve estimate
 - Note: the MSE from Benktander and Neuhaus are close to the optimal credibility MSE

Optimal Credibility Weights

Hürlimann derives the optimal credibility weights in sections 4-6. Many of the formulas are intermediary formulas in the derivation. For the exam, I would focus primarily on the final, simplified optimal credibility weight as well as the generalized optimal credibility formula (see the “Optimal Credibility Weights” recipe).

If we assume that the variance of the ultimate loss is the same as the variance of the burning cost ultimate loss estimate, $\text{Var}(U_i) = \text{Var}(U_i^{BC})$, we get the simplified optimal credibility weight formula. If we make a different assumption, then you need to use the generalized version of the formula.

You should definitely know the simplified optimal credibility weight formula. A potential twist to a question would be to use a different assumption, such as $\text{Var}(U_i) = 2 \times \text{Var}(U_i^{BC})$, and then use the generalized formula to calculate the optimal credibility weight.

Application to Standard Approaches

Hürlimann derived the optimal credibility weight formula for the loss ratio claims reserve approach. It can also be used with a more traditional approach, using LDFs to calculate the payout pattern (p_i^{CL}). With the LDF-based payout pattern, you can then calculate the reserve estimate as a credibility-weighting of the Chain Ladder and Cape Cod (or BF) reserve estimates using the Benktander, Neuhaus or optimal credibility weights.

Credibility-Weighted Cape Cod Approach

- Use LDFs to calculate the payout pattern (p_i^{CL})
- Calculate the ELR using the Cape Cod method

Credibility-Weighted Bornhuetter Ferguson Approach

- Use LDFs to calculate the payout pattern (p_i^{CL})
- The ELR is an assumption

Recipes for Calculation Problems

- Credible Loss Ratio Claims Reserve
- Optimal Credibility Weights

Overview

Brosius introduces the Least Squares method for estimating loss reserves and compares this method to the traditional Chain Ladder and the Budgeted Loss (Expected Loss) methods. The key theme of this paper is that the Least Squares method is a credibility weighting of the Link Ratio (Chain Ladder) and Budgeted Loss methods.

Least Squares Method

The Least Squares method fits a regression line through the data to estimate developed losses ($\hat{y}$).

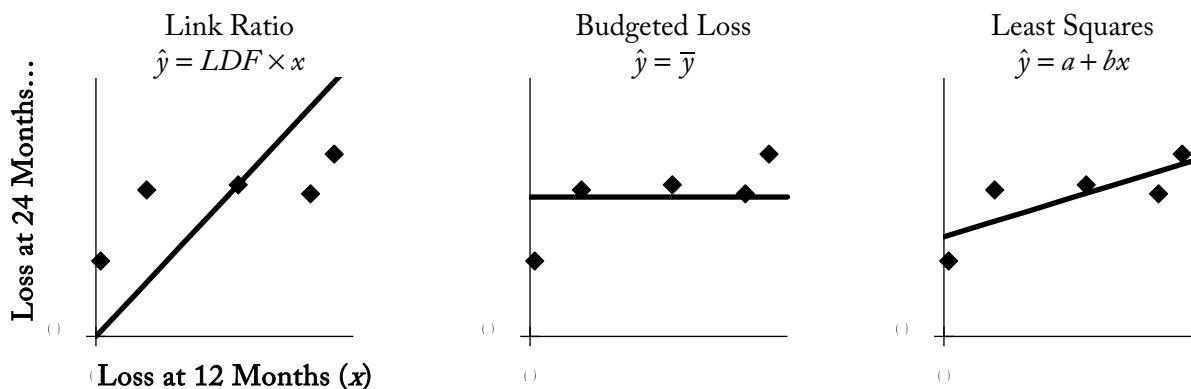
$$b = \frac{\overline{xy} - \bar{x} \cdot \bar{y}}{\overline{x^2} - \bar{x}^2}$$

$$a = \bar{y} - b \cdot \bar{x}$$

$$\hat{y} = a + bx$$

See the “Brosius – Least Squares” recipe.

Comparison of Methods



Least Squares as Credibility Weighting

The Least Squares method is a credibility weighting of the Link Ratio and Budgeted Loss methods. The Link Ratio and Budgeted Loss methods represent the extremes:

- Link Ratio: Places 100% credibility on loss experience and 0% on expected losses
- Budgeted Loss: Places 0% credibility on loss experience and 100% on expected losses

Least Squares Credibility Formula

The Least Squares method is flexible and places more (or less) credibility on the loss experience as appropriate.

Below are the key formulas for calculating the credibility on the link ratio method. The factor c is just the LDF for the link ratio method. The credibility is the ratio of b to the LDF. The closer b is to the LDF, the higher the credibility weighting the least squares method places on the link ratio method.

$$c = LDF = \frac{\bar{y}}{\bar{x}}$$

$$Z = \frac{b}{c}$$

Credibility-weighted formula:

$$\hat{y} = Z \times \frac{x}{d} + (1 - Z) \times E[y]$$

$$\hat{y} = Z \times LDF \times x + (1 - Z) \times E[y]$$

Special Cases

- If x and y are completely uncorrelated, then $b = 0$, resulting in the Budgeted Loss method where $\hat{y} = a$.
- If the regression line fits through the origin, then $a = 0$, resulting in the Chain Ladder method where $\hat{y} = bx$.
- The Bornhuetter-Ferguson method is a special case of Least Squares where $b = 1$. The BF method can be problematic if negative loss development is expected. The Least Squares method would allow b to adapt to the observed data.

Potential Problems (and Fixes) with Least Squares

The intercept is negative ($a < 0$):

- This causes the estimate of developed losses ($\hat{y}$) to be negative for small values of x .
- Solution: Use the link ratio method instead.

The slope is negative ($b < 0$):

- This causes the estimate of y to decrease as x increases
- Solution: Use the budgeted loss method instead

Key Assumptions for Least Squares

Least Squares assumes a steady distribution of random variables X and Y

- Least Squares is inappropriate if there's a systematic shift in the book of business.

Advantages of Least Squares

- Least Squares is more flexible than the link ratio, budgeted loss, and BF methods.
- Least Squares is a credibility weighting of the link ratio and budgeted loss estimates. It gives more (or less) credibility to the loss experience (x) as appropriate.

- Least Squares produces more reasonable results when the data has severe random, year-to-year fluctuations (e.g. a small book of business or thin data).

Adjustments to the data when using Least Squares

- When using incurred loss data, the data should be adjusted for inflation so that all accident years are on a constant-dollar basis.
- If there is significant growth in the book of business, you should divide the data by an exposure basis.

Hugh White's Question

If reported losses(x) come in higher than expected, the different methods will estimate different changes to the outstanding loss reserve:

- **Budgeted Loss Method (fixed prior case)** – The ultimate loss estimate is fixed, so we decrease the loss reserve estimate by the same amount as the unexpected increase in reported losses. This method treats the increased loss as losses coming in faster than expected.
- **BF Method** – The ultimate loss estimate increases by the amount losses were greater than expected. The loss reserve is unchanged. The BF method treats the unexpected increased loss as a random fluctuation (e.g. a large loss).
- **Link Ratio Method (fixed reporting case)** – The ultimate loss estimate increases in proportion to the excess losses by applying the LDF, so we increase the loss reserve estimate. This method assumes that a fixed percentage of ultimate losses is reported, so if reported losses increases, the ultimate loss estimate will increase proportionally.

Theoretical Models – Testing Least Squares

The purpose of this section is to test the least squares model against a few different theoretical loss models. With a theoretical model, we can use Bayes' Theorem to calculate the "correct" loss model and then see whether the least squares, budgeted loss or link ratio models have the same form as the Bayesian approach.

Model	Form	Model Constraints
Least Squares	$\hat{y} = a + bx$	
Link Ratio	$\hat{y} = bx$	$a = 0$
Budgeted Loss	$\hat{y} = a$	$b = 0$
Bornhuetter-Ferguson	$\hat{y} = a + x$	$b = 1$

Simple Model

- The number of ultimate claims incurred (Y) is either 0 or 1 with equal probability
- If there is a claim ($Y = 1$), there is a 50% chance it's reported by year end (X)

Using Bayes' Theorem, the best estimate of ultimate claims given x is $\hat{y} = \frac{1}{3} + \frac{2}{3}x$. This is the form $\hat{y} = a + bx$, so only the Least Squares method is compatible.

Poisson - Binomial Model

- The number of ultimate claims incurred (Y) is Poisson with mean μ
- Any given claim has probability d of being reported by year end

Using Bayes' Theorem, the best estimate of ultimate claims is $\hat{y} = x + \mu(1 - d)$. This is the same form as both the Least Squares method and BF method, since $b = 1$.

Negative Binomial – Binomial Model

- The number of ultimate claims incurred (Y) is Negative Binomial with parameters (r, p)
- Any given claim has probability d of being reported by year end

This model also has a Bayesian estimate with the same form as the Least Squares method, but the other methods will be incorrect.

Linear Approximation (Bayesian Credibility Approach)

We can only calculate the true Bayesian estimate by assuming a distribution for Y and $X|Y$, but that's not practical. Instead, we're going to find the best linear approximation to the Bayesian estimate of ultimate losses with Bayesian credibility, $L(x)$.

$$L(x) = (x - E[X]) \frac{\text{Cov}(X, Y)}{\text{Var}(X)} + E[Y]$$

Below is how a large reported loss (increasing x) can change the loss reserves, corresponding with the three different answers to Hugh White's question. For $x > E[X]$:

- $\text{Cov}(X, Y) < \text{Var}(X)$: loss reserve decreases
- $\text{Cov}(X, Y) = \text{Var}(X)$: loss reserve unaffected (ultimate loss increases by the increase to x)
- $\text{Cov}(X, Y) > \text{Var}(X)$: loss reserve increases

Using loss data, we can estimate $\text{Cov}(X, Y)$, $\text{Var}(X)$, and $E[Y]$, which gets us right back to the Least Squares method.

$$L(x) = \hat{y} = (x - \bar{x}) \frac{\overline{xy} - \bar{x} \cdot \bar{y}}{\overline{x^2} - \bar{x}^2} + \bar{y}$$

The key point is that the least squares method is the best linear approximation to the Bayesian estimate, although there will be sampling error in the parameter estimates of a and b .

Using simulated data from the Poisson-Binomial model, the Least Squares method fits the data better than the link ratio method and has a lower MSE.

Bayesian Credibility

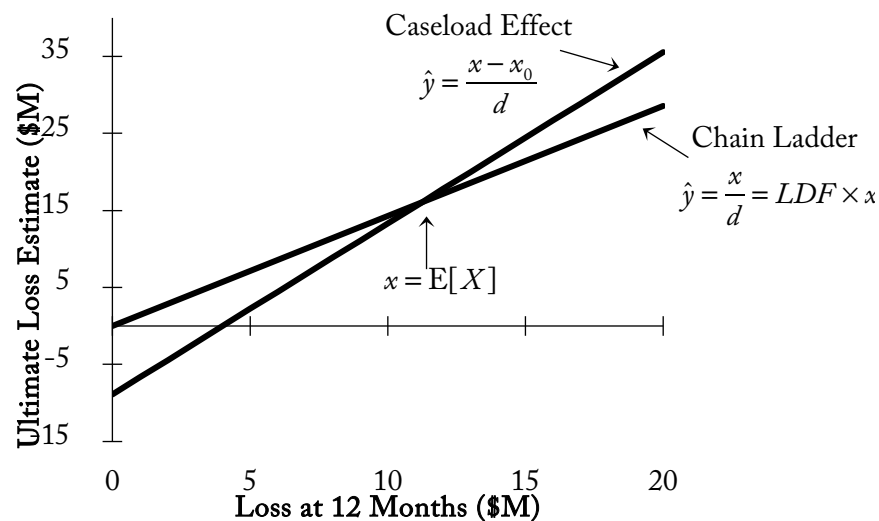
If the book of business changes significantly, we can't use the regular Least Squares method. But, if we make a few assumptions about the expected ultimate losses (Y) and the percent reported ($\frac{x}{Y}$), then we can calculate a Bayesian credibility estimate of ultimate losses (See the "Brosius – Bayesian Credibility" recipe).

Caseload Effect

The regular Bayesian credibility formula assumes that the expected percent of losses reported is the same no matter how large ultimate loss (Y) is. The caseload effect says that if ultimate loss is higher, then we would expect a lower percent of losses to be reported at time x (See the "Brosius – Caseload Effect" recipe).

Bayesian credibility still works, but the Bayesian credibility formula needs to be modified. Instead of using a fixed percent reported, the expected percent reported is lower if ultimate losses (Y) are higher. Below is a graphical view of the caseload effect and how the caseload effect estimate compares to the unmodified chain ladder estimate.

- If $x > E[X]$, the caseload estimate will be higher than the unmodified chain ladder estimate.
- If $x < E[X]$, the caseload estimate will be lower than the unmodified chain ladder estimate.



Recipes for Calculation Problems

- Least Squares Method
- Bayesian Credibility
- Caseload Effect

Overview

Below are some of the key topics to understand from the Friedland paper:

- The types and functions of reinsurance
- Differences in data and reserving between reinsurers and primary insurers
- The comparison of the volatility in development factors and patterns between reinsurance, primary insurance, and more
- How different reinsurance contracts interact and how to calculate ceded loss reserves

Appropriate reserving for reinsurance is important for the following reinsurer stakeholders:

- **Internal Management** – Sound reserves affect all areas of reinsurer operations (pricing, underwriting, strategic planning, financial decision-making, ...).
- **Investors** – Appropriately stated reserves are essential so that investors can properly evaluate the reinsurer's balance sheets and income statements for their decision-making.
- **Insurance Regulators** – Regulators rely on the financial statements of reinsurers to properly supervise the reinsurance market.
- **Rating Agencies** – If a reinsurer reports significant adverse reserve developments over time that reduce capital leaving the reinsurer in a weakened position, it could face a rating downgrade.

Functions of Reinsurance

- **Promote Stability** – Helps a ceding company stabilize loss experience over time and protect the ceding company from large unforeseeable losses. This can decrease the probability of ruin.
- **Increase Capacity** – By ceding a portion of all policies or its larger policies, a ceding company can increase its capacity to write more business, particularly at higher policy limits.
- **Protect against Catastrophes** – Reinsurance can protect ceding companies from catastrophic loss events as well as protect against casualty loss occurrences with multiple insureds (like terrorism).
- **Manage Capital and Solvency Margin** –
 - Reinsurance can help a ceding company pass the risk of large losses to the reinsurer which frees up capital since less capital will be required to support the policies written.
 - The ceded commission acts as a transfer of statutory surplus from the reinsurer to the cedent, which can manage financial results.
 - Premium ceded also reduces the cedent's net premium-to-surplus ratio (solvency margin), which allows the cedent to write more business.

- **Access Technical Expertise** – Reinsurers have technical expertise in underwriting, marketing, claims, loss prevention, pricing, and entering new lines/regions that can help small insurers.
- **Other Functions** – Reinsurance can help a ceding company withdraw from a line of business, geographic area, or production source.

Types of Reinsurance

Reinsurance is categorized as **Treaty** or **Facultative** and **Proportional** or **Non-Proportional**.

Treaty Reinsurance

- The ceding company cedes all business arising from the lines of business that fall within the terms of the treaty subject to treaty limits.
- There is no underwriting by the reinsurer of individual risks within the treaty terms.
- The cedent has an obligation to cede a risk under the treaty terms and the reinsurer has an obligation to automatically accept it.

Facultative Reinsurance

- Both the ceding company and reinsurer have the option (faculty) to accept or reject individual submissions and an individual reinsurance agreement is negotiated for each policy ceded.
- Primarily used to increase capacity, typically for high-value and hazardous commercial risks.

Proportional Reinsurance

- Increases capacity and manages capital and solvency margins to provide surplus relief
- Both premiums and losses are shared between the cedent and reinsurer based on the cession percentage. The reinsurer pays a **ceding commission** to reimburse the cedent for expenses underlying the policy.

Types of Proportional Reinsurance:

- **Quota Share** – The ceding company cedes an agreed percentage of each risk (premiums and losses) to the reinsurer for the lines of business in the contract, typically a treaty.
- **Surplus Share** – The cedent cedes the surplus amount of a risk above the retained line subject to a maximum ceded percentage and limit. The proportion ceded is different for different insured policies according to the underlying policy limits and acts like a variable quota share.

$$\text{Proportion Ceded} = \frac{\text{Policy Limit} - \text{Retained Line}}{\text{Policy Limit}}$$

Non-Proportional Reinsurance (Excess of Loss)

- Provides stability by protecting losses above a limit for risks ceded
- Loss ceded is based on the size of loss and the premium is not proportional to the coverage limits

Types of Non-Proportional Reinsurance:

- **Excess per Risk** – Excess-of-loss reinsurance above a retention for *each risk*
 - Primarily to protect property exposures (e.g. 7M excess 3M on property policies with 10M policy limits)
 - Allows ceding companies to write larger risks (increase capacity)
- **Excess per Occurrence and Catastrophe** – For a loss occurrence, the ceding company retains the retention and cedes the loss excess the retention to the reinsurer up to the reinsurance limit.
 - **Excess per Occurrence** – Protects a cedent from the accumulation of losses in a single loss occurrence.
 - **Catastrophe Reinsurance** – Form of Excess per Occurrence for a single catastrophic event or series of events. Most allow for reinstatement after a full limit loss.
- **Annual Aggregate Excess of Loss (stop-loss)** – Guarantees a ceding company's losses won't exceed a predetermined threshold (percent of premium or fixed dollar amount)
 - The reinsurer indemnifies for losses above the aggregate value
 - Protects net results (other reinsurance inures to the benefit of the Agg. Excess of Loss)
 - The best option to protect capital but can be very expensive or unavailable
- **Clash** – Casualty reinsurance contract that attaches above all other policy limits. It protects a ceding company when there are multiple claims from multiple insureds for the same catastrophe and its reinsurance policy doesn't fully reimburse the insurer.
 - Components of a clash event:
 - Loss must have multiple policies by one insured or similar policies held by multiple insureds
 - The losses are traceable to and the direct consequence of a specific event
 - The event must take place within a specific timeframe

Finite Risk Reinsurance

This type of reinsurance takes the time value of money into account. Features include:

- Risk transfer and risk financing in a multi-year contract
- Incorporates the time value of money and investment income
- Limited assumption of risk by the reinsurer
- Reinsurer and ceding company share results

Includes run-off solutions, which transfer reserve development risk to the reinsurer. Reasons for run-off include corporate restructuring, mergers & acquisitions, discontinuation of lines of business, erratic changes in the valuation/cost of a liability, ...

Loss Portfolio Transfers

- Transfers all (or a portion) of liability for future loss payments on losses already incurred

- Relieves cedent of uncertainty in loss reserves and relieves capital

Adverse Development Cover

- Ceding company is reimbursed for losses excess a retention, but there's no transfer of the loss reserves to the reinsurer
- Often used for Mergers & Acquisitions to transfer the risks of timing and adverse reserve development

Reinsurance Concepts and Contract Provisions

Inure to the Benefit of

The concept of “inuring to the benefit of” specifies whether a treaty takes effect to the benefit of the reinsurer or the reinsured.

For a reinsurance Treaty A:

- If other reinsurances apply first to reduce the loss subject to Treaty A, then the other reinsurance contracts *inure to the benefit of the reinsurer* of Treaty A.
- If other reinsurances are ignored with respect to Treaty A (they don't lower the loss subject to Treaty A), then the other reinsurance contracts *inure to the benefit of the ceding company*.

Losses-Occurring During vs Risks-Attaching

- **Losses-Occurring During** – Reinsurance coverage for all losses that occur between the inception and expiration of the reinsurance contract, regardless of when the underlying policy was issued.
- **Risks-Attaching** – Reinsurance coverage for losses on underlying policies with inception dates during the reinsurance contract's effective period. The underlying policies “attach” to the contract.

Subscription Percentage

A subscription policy is a reinsurance policy where risk is shared by multiple reinsurers. Each reinsurer has a subscription percentage to the contract.

Reasons for this approach include:

- When coverage is more than one reinsurer is willing to assume
- Allows the cedent to diversify credit risk (the risk that a reinsurer can't pay reinsurance recoveries)

Commutation Clause

A commutation is the cancellation of a reinsurance contract. The reinsurer pays the present value of reinsurance recoveries not yet due for the termination of the contract and all future obligations.

Both reinsurers and ceding companies have reasons for commutations:

For a Ceding Company

- To exit a line of business or region
- To manage reserves for transfer or sale
- To avoid credit risk of a reinsurer
- To better manage claims and claims-related expenses

For a Reinsurer

- To terminate a relationship with a ceding company
- To protect itself from the insolvency of a ceding company
- To avoid disputes with the cedent about future loss development

Sufficient and Reliable Data

Sufficiency

Data are sufficient if they included all information needed for the actuarial work.

For the development method:

- Reinsurance data may not be appropriate for the underlying development method assumptions because of its manuscript nature (custom-written) and due to operational changes at ceding companies, the reinsurer or both
- Changes at the ceding company level can violate the assumption of consistency in mix of business, attachment points, limits, claims processing, etc.

Reliability

Data are reliable if they are sufficiently complete, consistent, and accurate for the purposes of the work.

- Data should be validated: Reviewed for consistency, completeness, and accuracy.

Reliability challenges for reinsurers vs. primary insurers:

- Each ceding company/broker has different IT systems, terminology, etc.
- Bordereau reporting can differ (types of data, labeling, frequency)
- Reporting lags
 - Claims are reported first to ceding company before being reported to the reinsurer
 - Long-tailed nature of certain reinsurance like excess per risk and catastrophe reinsurance
 - Bordereau reporting: losses are only reported quarterly or more infrequent schedule
- Gaps in reporting of critical claims and claims expense information by the ceding company
- Manuscript nature of reinsurance policies

Homogeneity and Credibility of Data

Homogeneity

Homogeneous Risk Group (HRG) – Set of (re)insurance obligations with similar risk characteristics to allow for reliable valuation of unpaid losses or technical provisions.

Data should be segmented into groups with similar characteristics of loss experience such as consistency of coverage, similar reporting/payment patterns, ability to develop appropriate case outstanding for claims, severity, and volume of losses in the group.

Credibility

Credibility – A measure of the predictive value for a set of data.

Credibility for a homogeneous risk group **increases** with:

- **Increasing** homogeneity of the data within a group
- **Increasing** the volume of data in a group

If groups are too granularly defined, the volume of data in each group may be too low for a reliable analysis.

Important Variables for Partitioning the Reinsurance Portfolio

- **Line of business** (property, casualty, ...)
- **Type of contract** (facultative, treaty, ...)
- **Type of reinsurance cover** (quota share, excess per-occurrence, CAT, ...)
- **Primary line of business** – for casualty
- **Attachment point** – for casualty
- **Contract terms** (flat-rated, retro-rated, claims-made, ...)
- **Type of cedent** (small, large, ...)
- **Intermediary**

Organization of Data by Experience Period

Accident Year Aggregation

Losses are grouped by date of occurrence and calendar year earned premium is used to approximately match accident year losses.

Advantages

- Easy to achieve and understand
- Losses over shorter time frame than underwriting year so losses can be reliably estimated sooner
- Industry benchmarks are based on AY
- Valuable when there are economic/regulatory changes or major loss events

Disadvantages

- Potential mismatch of losses and premiums

Underwriting Year Aggregation

Losses are grouped by the year in which the reinsurance policy became effective (inception date).

<u>Advantages</u>	<u>Disadvantages</u>
• True match of losses and premiums • Valuable when underwriting or pricing changes occur	• Extended timeframe for losses tied to UW year • Difficult to isolate the effect of a single large loss event

Types of Data

ULAE – Usually excluded from reinsurance coverage.

ALAE – Generally three possible reinsurance treatments:

- Included with claim amount to determine excess of loss coverage
- Included on a pro-rata basis (% of reinsured loss / total loss)
- Not included in coverage

Multiple currencies – Loss data may be in different currencies. There are two approaches:

- Separate data by currency, then combine the data after translating to a common currency using the exchange rate at a single point in time
- Aggregate losses are based on the ceding company's currency of origin

Large losses – Exclude large losses from initial projections and then add in case-specific projections of the reported portion for large losses and a smoothed provision for the IBNR portion of the large losses.

Recoveries (deductibles, salvage & subrogation) – Generally recoveries are applied before reinsurance.

Data Challenges for Reinsurers

- **Influence of Change in Operations and Environment**
 - Operational changes for ceding insurers and reinsurers
 - Changes in the legal/economic environment of ceding companies that impact losses
- **Other Experience Typically Excluded**
 - Actuary may exclude discontinued business (run-off) from the data analysis
- **Reporting Lags**
 - Losses first reported to ceding company before they're reported to the reinsurer
 - Losses may not be reported to the reinsurer until a claim hits a certain threshold
- **Heterogeneity of Contract Wording**
 - The “manuscript nature” of reinsurance contracts means contract wording can be different by contract making it more difficult to aggregate similar data

External Sources of Data

Actuaries at smaller reinsurers often use external data sources to help analyze development or tail factors, trend rates, expected loss ratios, etc. Below are some external sources of data:

- Reinsurance Association of America (RAA)
- Best's Aggregates & Averages
- Reports from global brokers or reinsurers
- Other internet searches

External data may be misleading or irrelevant due to differences in:

- Manuscript wording/term of reinsurance contracts
- Mix of assumed business (differences in industry, region, attachment points, policy limits)
- Types of reinsurance (treaty vs. facultative, proportional vs. non-proportional)
- Underwriting processes
- Claims management differences
- Coding and IT system differences

Methods Used

Development Method

- Assumes future development is like prior years' development and losses to date are predictive of losses yet to be observed
- Assumes consistency across experience period of claims processing, mix of business, policy limits, reinsurance coverage, etc.

Expected Method

Applies an expected loss ratio to earned premium to estimate expected losses.

Often used when:

- Entering a new line of business or region
- When historical losses are irrelevant to project future losses due to changes in strategy, operations, or environment
- When the development method isn't appropriate for immature periods because development factors are too highly leveraged
- When data are unavailable for other methods

Bornhuetter-Ferguson Method

- Assumes unreported (or unpaid) loss will develop based on expected loss.
- Relies on both a selected loss development pattern and expected loss estimate

Differences in Method Assumptions for Reinsurance vs Primary Insurance

- For a similar line: **LDFs at immature ages are often higher** (more leveraged) for reinsurance due to reporting lags
- Loss trend factors are often **higher for excess of loss reinsurance** than primary insurance
- Less precision in premium on-level factors for rate changes for reinsurance vs. primary insurance
- Limited use of adjustment factors for tort/product reform for reinsurance vs primary insurance

Effect of Changes in Currency Exchange Rates

Many global reinsurers review triangles **at the prevailing** exchange rates.

- This prevents distortions in age-to-age factors due to fluctuating exchange rates year-to-year.

Comparisons of Development Factors and Patterns

Reinsurance vs. Primary Insurance (Similar Type of Business)

More volatility in age-to-age factors at earlier maturities for:

- Reinsurance compared to primary insurance
- Paid losses compared to reported losses

Greater volatility means more uncertainty in age-to-age factor selection and projected ultimate losses.

Longer reporting/payment patterns for reinsurance due to lags in reporting to the reinsurer.

Proportional vs. Non-Proportional Reinsurance (Same Line of Business)

For the same line of business:

- Significantly more volatility in age-to-age factors for non-proportional treaty and facultative reinsurance than for proportional treaty reinsurance
- CDFs are greater for non-proportional treaty and facultative reinsurance (longer dev. patterns)
- Also - More volatility in ratios of paid-to-reported losses for non-proportional and facultative

A key difference is that proportional reinsurance is ground-up and non-proportional is excess of loss.

Takeaway: The greater volatility in age-to-age factors and diagnostics results in greater uncertainty in projected ultimate loss from the development method.

Property Reinsurance excl. Catastrophe vs. Property Reinsurance Catastrophe

- Reinsurer carried reserves for catastrophe losses are usually based on ground-up exposure-based assessments using info from ceding companies by contract (NOT the development method)
- The impact of catastrophe events at different times of the year impacts development and age-to-age factors for different years

- Development method assumptions may not be appropriate
- Volatility is much higher for catastrophe vs. ex-catastrophe property reinsurance.

Takeaway: Methods using age-to-age factors are often not appropriate for property catastrophe.

Implications of Volatility in Loss Development

- Greater volatility in age-to-age factors can lead to greater uncertainty in projections of ultimate loss (and therefore unpaid loss) because:
 - Development methods and other methods rely on the age-to-age factors that have greater volatility (B-F method does and the expected method often does to set ELRs)
 - Greater volatility in indicated ultimate loss ratios is often used to select ELRs for the expected and B-F methods

Takeaway: Greater volatility in projected ultimate loss results in a greater range in indicated IBNR and Total Unpaid for:

- Reinsurance compared to primary insurance
- Non-proportional treaty and facultative compared to proportional treaty reinsurance
- Catastrophe reinsurance compared to excluding catastrophe reinsurance for property

Quota Share and Stop Loss Reinsurance Examples

Quota Share Reinsurance

- Apply the %ceded (or $1 - \%retained$) to gross ultimate loss, case reserves, paid loss, and IBNR to get the ceded values.
- The %ceded may change over time, so calculations are applied by year

Stop-Loss Reinsurance

- Stop-Loss reinsurance applies after all other reinsurance and protects the net result of a ceding company
- Once a ceding company breaches stop-loss coverage, the reinsurer will often increase the price or the attachment point (or both)

LDF Curve-Fitting and Stochastic Reserving

Overview

Clark is a calculation-heavy paper and you should be able to do the Variance of Reserve calculations effortlessly by the time of the exam, a problem which comes up regularly. You should also be prepared for questions about model assumptions, diagnostic graphs and how to evaluate results for reasonableness.

The focus of Clark is to create develop a loss reserving model that estimates a central loss reserve estimate (with the LDF or Cape Cod method) and can calculate the variance of the reserve estimate.

Goals of a loss reserving model

- Mathematically describe loss emergence to estimate loss reserves
- Estimate the reserve range around the expected reserve, due to variance from:
 - Process variance - uncertainty due to randomness
 - Parameter variance - uncertainty in expected value

Expected Loss Emergence

Instead of using LDFs directly, we're going to fit a curve, $G(x)$, to incremental losses using MLE in order to get the best-fitting parameters. Then, we'll use this curve to estimate the payment pattern (from 0% to 100%) as an accident year matures.

Clark uses two different curves, the Loglogistic and Weibull curve. The Weibull curve has a lighter tail.

Loglogistic

$$G(x) = \frac{x^\omega}{x^\omega + \theta^\omega}$$

Weibull

$$G(x) = 1 - e^{-\left(\frac{x}{\theta}\right)^\omega}$$

x - average age of loss occurrence.

- Advantages of using a parameterized curve for the loss emergence pattern
- Estimating unpaid losses is simplified (only need 2 parameters)
- Can also use data that's not from a triangle with evenly spaced evaluation dates
- Payout pattern, $G(x)$, is a smooth curve and doesn't overfit like age-to-age factors might

Advantage of using data in a tabular format

- Can use data at irregular evaluation periods or when you only don't have the full triangle

Process Variance

Process variance is the variance due to randomness in the insurance process.

Loss Model Assumptions

- Assume incremental losses have a constant variance/mean ratio, σ^2
- Assume incremental losses follow an over-dispersed Poisson model

Fitting the loss emergence curve

We find the best-fitting curve using the maximum likelihood method. For a given set of parameters, we calculate the expected incremental losses, μ_i , for each cell of the loss triangle. Then, we compare the likelihood that the actual incremental losses came from an over-dispersed Poisson distribution with the parameters μ_i and σ^2 .

The goal is to find the set of parameters that maximize the log-likelihood function. See the “Finding Best-Fit Parameters with MLE” recipe for an example of how to do this.

Advantages of using an over-dispersed Poisson distribution to model incremental losses:

- Using a scaling factor, σ^2 , allows us to match the 1st and 2nd moments of other distributions
- The Maximum Likelihood Estimate reproduces LDF and Cape Cod loss reserve estimates

Parameter Variance

Parameter variance is the variance in the estimate of the parameters. It's calculated based on the Rao-Cramer lower-bound approximation, using the second derivative information matrix. The information matrix is used to calculate the covariance matrix, which is used to calculate the parameter variance. The actual calculation of the parameter variance is too complex for the exam.

Key Model Assumptions

- **Incremental losses are iid**
 - Independent – One period doesn't impact surrounding periods (this assumption fails if there are calendar year effects such as inflation)
 - Identically distributed – Assume the same emergence pattern, $G(x)$, for all accident years (this assumption fails if the mix of business or claims handling changes)
- **Variance/Mean Scale parameter, σ^2 , is fixed and known**
 - We ignore the variance of σ^2
- **Variance estimates use approximation to the Rao-Cramer lower bound**

The model assumptions mean there's a potential that future losses have higher variance than what the model indicates.

LDF Method

A problem with the LDF method is that there is a parameter for each accident year for the current $Loss_{AY}$. See the “Variance of Reserves (LDF Method)” recipe for how to calculate reserve variance.

Cape Cod Method

The Cape Cod method uses additional information, an exposure base. Clark recommends on-level earned premium, but another exposure base can be used as long as it's proportional to ultimate expected losses by accident year.

Premium must be on-leveled so that we can assume a constant ELR across all accident years. We could also adjust for loss trend net of exposure trend so that all accident years are at the same cost level.

LDF Method vs. Cape Cod Method

- LDF method over-parameterizes the model, fits to the “noise” in the data
 - For a 10-yr triangle, there are 12 parameters to estimate and only 55 data points
- Cape Cod has lower parameter variance because there are fewer parameters to estimate and it uses more information (on-level premium)

Variance of Loss Reserves

Once you have the loss reserve estimate, the scale parameter and the parameter variance, you can calculate the variance around the loss reserve estimate.

$$ProcessVar = \sigma^2 \cdot Resv$$

$$TotalVariance = ProcessVariance + ParameterVariance$$

Diagnostics

ELR for Cape Cod

The Cape Cod method assumes a constant ELR across all accident years. Test this assumption by graphing the estimated ultimate loss ratios by accident year.

The estimated ultimate loss ratios should be random around the ELR with no patterns or trends. If there is a pattern, then the assumption of a constant ELR isn't reasonable and the Cape Cod reserve estimate could be biased.

$$\overline{Ult Loss Ratio}_{AY} = \frac{Loss_{AY}}{Prem_{AY} \times G(x)}$$

Residual Graphs

$$r_{AY,k} = \frac{IncLoss_{AY,k} - \mu_{AY,k}}{\sqrt{\sigma^2 \cdot \mu_{AY,k}}}$$

$$Norm.residual = \frac{actual\ incremental - expected\ incremental}{\sqrt{\sigma^2 \cdot expected\ incremental}}$$

Residual graphs are an important diagnostic to test the underlying assumptions. Below are the graphs Clark specifically mentions and what assumptions they test:

- **Normalized Residuals vs. Increment Age**
 - Tests how well the loss emergence curve fits incremental losses at different development periods.
- **Normalized Residuals vs. Expected Incremental Loss (μ_i)**
 - Tests the variance/mean ratio σ^2 : If the variance/mean ratio is not constant, we should see residuals clustered closer to zero at either high or low expected incremental losses.
- **Normalized Residuals vs. Calendar Year**
 - Tests whether there are diagonal effects (e.g. high inflation in a calendar year)

For all the graphs, the residuals should be random around zero with no patterns or autocorrelations. If this isn't the case, then some assumptions of the model are incorrect.

Process vs. Parameter Variance

We should see that parameter variance is greater than process variance. This is because most of the uncertainty is due to the inability to estimate the expected reserve (parameter variance) rather than uncertainty due to random events (process variance).

The reason for this is that there are so few data points in a loss triangle to estimate the parameters. The Cape Cod method lowers the parameter variance by including the exposure data.

Other Calculations

Variance of Prospective Losses

The regular Clark method calculates reserve variance for past accident years. This same approach can be used to calculate variability around prospective losses for the prospective accident year (period). See the "Variance of Prospective Losses" recipe for this calculation.

Calendar Year Development

Instead of estimating the total unpaid loss reserve, we can calculate the unpaid losses that we expect will be paid over the next calendar year and create a range around that.

The advantage of this type of calculation is that the model can be tested in a relatively short period of time. After one year, the actual 12-month loss development can be compared to the original forecasted range to test whether the actual development falls within the range. See the "Variance of Calendar Year Development" recipe for this calculation.

Variance of Discounted Reserves

The discounted paid loss reserve is calculated by discounting the future payments at the half-year mark. For the exam a calculation problem doesn't seem testable.

One key point is that the CV of the discounted loss reserve is smaller than the CV of the undiscounted loss reserve. This is because the tail of the payout curve has the greatest parameter variance, but is discounted the most.

Adjustments for Other Exposure Periods

The formula for $G(x)$ is only valid on an accident year basis where the first development period is at 12 months. If the first development period is shorter than 12 months or policy year is used, then you need to make some adjustments.

Percent of ultimate loss as of time t , with annualization:

$$G(t) = \text{expos}(t) \cdot G(x) \quad \text{where} \quad x = \text{AvgAge}(t)$$

	Accident Year	Policy Year
$\text{expos}(t)$	$= \begin{cases} \frac{t}{12} & t \leq 12 \\ 1 & t > 12 \end{cases}$	$= \begin{cases} \frac{1}{2} \cdot \left(\frac{t}{12}\right)^2 & t \leq 12, \text{ note: } \text{expos}(12) = 50\% \\ 1 - \frac{1}{2} \left(2 - \frac{t}{12}\right)^2 & 12 < t \leq 24 \\ 1 & t > 24 \end{cases}$
$\text{AvgAge}(t)$	$= \begin{cases} \frac{t}{2} & t \leq 12 \\ t - 6 & t > 12 \end{cases}$	$= \begin{cases} \frac{t}{3} & t \leq 12 \\ \frac{(t-12) + \frac{1}{3}(24-t)(1-\text{expos}(t))}{\text{expos}(t)} & 12 < t \leq 24 \\ t - 12 & t > 24 \end{cases}$

I would know all the formulas for accident year. For policy year, I would make sure to know the special cases where $t = 12$ months and where $t = 24$:

Policy Year	$\text{expos}(t)$	$\text{AvgAge}(t)$
$t = 12$	50%	4
$t = 24$	100%	12

Recipes for Calculation Problems

- Variance of Reserves (LDF Method)
- Variance of Reserves (Cape Cod Method)
- Normalized Residuals
- Variance of Prospective Losses
- Variance of Calendar Year Development
- Finding Best-Fit Parameters with MLE

Mack – Chain Ladder

Measuring the Variability of Chain Ladder Reserve Estimates

Overview

The goal of this paper is to use the chain ladder method to create a confidence interval around the ultimate loss and estimated loss reserve.

Mack - Chain Ladder focuses on two main ideas:

1. **Three main assumptions** underlie the chain ladder method. For a loss triangle, we should look at different diagnostic graphs and tests to see whether the chain ladder method is appropriate or not.
2. Based on the three assumptions, we can calculate the estimated loss reserve and the standard error of the estimated loss reserve for each accident year and for all years combined with the chain ladder method. Then, after making an assumption about the distribution of the loss reserve (e.g. lognormal), we can calculate loss reserve confidence intervals.

Mack Assumptions

1. Expected losses in the next development period are proportional to losses-to-date.

$$E[C_{i,k+1} | C_{i,1}, \dots, C_{i,k}] = C_{i,k} \cdot LDF$$

- The chain ladder method uses the same LDF for each accident year
- Uses most recent losses-to-date to project losses, ignoring losses as of earlier development periods

2. Losses are independent between accident years.

3. Variance of losses in the next development period is proportional to losses-to-date with proportionality constant α_k^2 that varies by age.

$$\text{Var}[C_{i,k+1} | C_{i,1}, \dots, C_{i,k}] = C_{i,k} \cdot \alpha_k^2$$

Alternative Variance Assumptions (Weighted LDFs)

The chain ladder method in Mack - Chain Ladder uses volume-weighted LDFs. This implies that variance of losses in the next development period ($Loss_{k+1}$) is proportional to losses-to-date ($Loss_k$), Assumption 3.

If we calculate the LDFs a different way, such as the simple average of the age-to-age factors or age-to-age factors weighted by $Loss_k^2$, then we're making a different variance assumption.

Variance Assumptions		
$\text{Var}(C_{k+1}) \propto$	LDF calc. Weight	LDF
1	$C_{i,k}^2$	$C_{i,k}^2$ -wtd
$C_{i,k}$	$C_{i,k}$	Vol-weighted
$C_{i,k}^2$	1	Simple Avg

Variance of Reserves

Using the three chain ladder assumptions, we can calculate the variance of the reserve estimate with Mack's formula for the standard error of the reserve. We get the estimated reserve and the standard error of the estimate by accident year and for the overall reserve.

One disadvantage of the Mack method (compared to the bootstrap method) is that it doesn't tell us about the shape of the loss reserve distribution. It only gives us the mean and standard deviation. To create confidence intervals, we need to make an assumption about the distribution.

Reserve Confidence Intervals

Once we've done all the calculations, we have an expected loss reserve estimate (the loss reserve estimate from the standard chain ladder method) and the standard error (standard deviation) of the loss reserve estimate. Besides these values, the method doesn't tell us anything else about the distribution of the loss reserve estimate.

So, we're going to assume a distribution and use the mean and standard deviation of the loss reserve that we calculated. Mack looks at two distributions: Normal and Lognormal.

Normal Distribution

$$\text{C.I.} = R \pm z \cdot \text{s.e.}(R)$$

- If min of C.I. is negative OR $\text{s.e.}(R) > 50\%$ of R , use Lognormal

Lognormal Distribution

$$\sigma^2 = \ln \left[1 + \left(\frac{\text{s.e.}(R)}{R} \right)^2 \right]$$

$$\text{C.I.} = R \cdot \exp \left[\pm z \cdot \sigma - \frac{\sigma^2}{2} \right]$$

Checking the Chain Ladder Assumptions

The three Mack assumptions have significant implications, so we should take a look at some different tests and diagnostics to see how well the assumptions hold up.

Plot of Cumulative Losses from Adjacent Periods

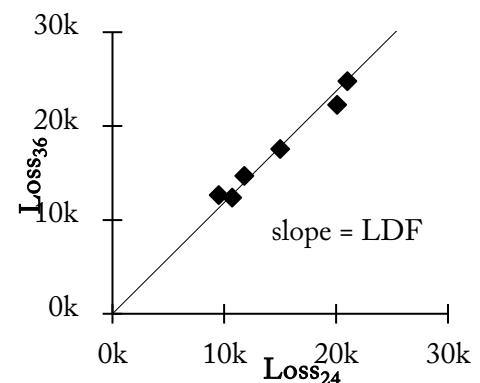
This plot tests assumption 1. We want to see if losses at the next period are proportional to losses-to-date with no intercept.

If the assumption holds, you should see:

- Linear relationship between $Loss_{k+1}$ and $Loss_k$ through the origin (no intercept)
- Line should go through the data points

If the assumption is violated, you might see:

- The data points show that there should be an intercept term in the regression line
- The relationship isn't linear

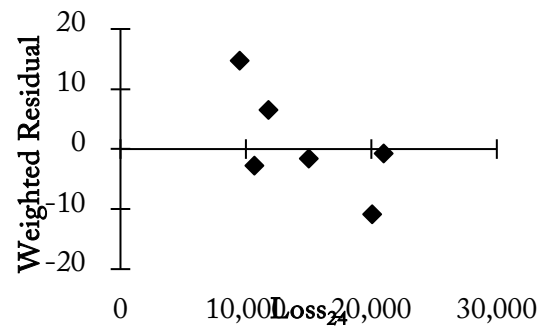


Plot of Weighted Residuals

This plot primarily tests assumption 3, the variance assumption.

If the assumption holds, you should see:

- Residuals should be random around zero with no significant trends or patterns.



If the assumption is violated, you might see:

- Variance of residuals is higher at one end of the graph than at the other
- Residuals show an increasing (or decreasing trend)

If the residuals for a few development periods aren't random, we can graph the residual plots using the LDFs for the alternative variance assumptions: $\text{Var}(C_{k+1}) \propto C_k^2$ and $\text{Var}(C_{k+1}) \propto 1$. If the residual plots for one of the alternative assumptions is more random, then we could replace the volume-weighted LDF with the alternative LDF (f_{k0} or f_{k2}).

Testing Assumption 2 – Independence between accident years (Calendar Year Test)

The assumption of independence between accident years implies that there are no calendar year effects that impact losses from multiple accident years (causing a dependence between accident years).

We test the null hypothesis that there are no calendar year effects with Mack's calendar year test (see the "Mack - Chain Ladder – Calendar Year Test" recipe). If there are significant calendar year effects, then assumption 2 is violated.

Examples of calendar year effects:

- Major changes in claims handling practices
- Major changes in setting case reserves
- Unexpectedly high (or low) inflation
- Significant changes due to court decisions

Testing for Correlation Between Adjacent LDFs (Assumption 1)

The first assumption implies that subsequent development factors are uncorrelated (e.g. high development from ages 12 to 24 gives no information about how losses develop from age 24 to 36).

In the chain ladder method, we always use the same LDF from ages 24 to 36 no matter how losses developed from ages 12 to 24. For instance, if the overall LDF from ages 12 to 24 is 3.5 and a particular AY developed from 1,000 to 5,500 (age-to-age factor of 5.5), we still expect the same % development from 24 to 36.

- **Example:** If a book of business typically shows a smaller-than-average increase, $\text{Loss}_{k+1}/\text{Loss}_k < \text{LDF}_k$, after a larger-than-average increase, $\text{Loss}_k/\text{Loss}_{k-1} > \text{LDF}_{k-1}$, then the chain ladder method would not be appropriate.

Mack - Chain Ladder uses Spearman's rank correlation to test for correlation between LDFs (See the "Mack - Chain Ladder – Correlation of Adjacent LDFs" recipe).

The Mack Correlation of Adjacent LDFs test looks at the triangle as a whole, not at each column-pair separately. This is because it's more important to know if correlation between LDFs prevails throughout the triangle.

Advantages of using Spearman's rank:

- The test is distribution-free (doesn't assume LDFs are from a normal distribution)
- Differences in variances of LDFs between development periods is less important because it uses ranks

Reviewing Results for Reasonableness

- MSE of the total reserve is greater than the sum of the MSE from the reserve for individual accident years.
 - Because the same LDFs are used for all accident years, the loss reserve estimates are positively correlated between accident years, increasing the total MSE.
- Higher standard error percentage for the oldest accident years
 - The absolute standard error should be lowest for the oldest accident years, but since the size of the loss reserve is so small, the standard error percentage will be higher.
- Higher standard error percentage for the most recent accident year (or two)
 - Uncertainty in forecasting future losses is highest for the most recent accident years because losses are immature, so the standard error percentage will be higher.

Weaknesses of the Chain Ladder method

- Estimators of the last 2 or 3 LDFs rely on very few observations
- Doesn't work well for the most recent accident year where losses-to-date provide a very uncertain base to project ultimate losses
 - Another method, such as the least squares method would put less credibility on the immature losses.

Recipes for Calculation Problems

- Residual Test
- Calendar Year Test
- Reserve Confidence Interval
- MSE Calculation
- Correlation of Adjacent LDFs
- Overall MSE Calculation

Venter Factors

Testing the Assumptions of Age-to-Age Factors

Overview

Think of Venter Factors as an extension of the Mack – Chain Ladder paper. Venter examines the three key Mack assumptions underlying the chain ladder method and identifies six ways to test those assumptions. If those tests fail for a loss triangle, Venter points out a few alternative loss reserving methods to use instead.

As you go through Venter Factors, think about how it ties together with Mack (1994) to give a more complete picture of the chain ladder method and how to use the chain ladder method to create reserve ranges.

Mack Assumptions – Restated

If the assumptions below hold, then the chain ladder method gives the minimum variance unbiased linear estimator of future loss emergence.

Assumption 1

Expected incremental loss emergence is proportional to cumulative losses-to-date

$$E[IncLoss_{AY,k+1} | data] = (LDF_k - 1) \cdot Loss_{AY,k}$$

Assumption 2

Losses are independent between accident years

Assumption 3

Variance of the next incremental loss is a function of age and cumulative losses-to-date

$$Var[IncLoss_{AY,k+1} | data] = function(k, Loss_{AY,k})$$

We can't directly test these assumptions, but there are six testable implications that fall out of these assumptions that we can test. If the tests for any of the implications fails, then the three Mack assumptions don't hold up, meaning that the chain ladder method doesn't give the minimum variance unbiased linear estimate of future losses.

Testable Implication 1 – Significance of development factors

- **Tests Assumption 1**: Linear loss emergence proportional to loss-to-date

Assumption 1 assumes incremental losses are proportional to losses-to-date with a development factor. If that's the case, the development factor should be statistically significant.

Later in the paper, Venter shows the regression including a constant, $\hat{y} = a + bx$. This tests the alternative linear with constant (least squares) model compared to the chain ladder model and also indirectly tests for implication 1. Excluding the constant term in the regression will give a more direct test of implication 1.

One last point: You can't just do a regression of $Loss_{k+1}$ vs. $Loss_k$. This is because what we really want to predict is the incremental loss from one period to the next. If we use $Loss_{k+1}$, then comparing the coefficient b to the standard deviation directly will overstate the significance of the factor.

Testable Implication 2 – Superiority to alternative loss emergence patterns

- **Tests Assumption 1:** Linear loss emergence proportional to loss-to-date

If assumption 1 is correct, then the chain ladder model should predict incremental losses better than alternative loss emergence patterns. We can use a goodness-of-fit test to compare the alternative models to the chain ladder method.

Different Loss Emergence Patterns

Expected Incremental Loss is...	Model
...proportional to loss-to-date	Chain Ladder
...proportional to loss-to-date with a constant	Least Squares
...percentage of ultimate loss	Bornhuetter-Ferguson: $f(d)h(w)$

Alternative Pattern 1 – Linear with a constant (least squares method)

$$E[IncLoss_d] = \underset{\text{factor}}{f(d)} \cdot Loss_k + \underset{\text{constant}}{g(d)}$$

Including a constant term is often significant in the first development period or two, especially for highly variable and long-tail lines.

Test this alternative pattern by running a linear regression of $IncLoss_{k+1}$ vs. $Loss_k$ to get the factors of the development formula $\hat{y} = a + bx$. If the constant is statistically significant, this loss emergence pattern is more strongly supported than the chain ladder method.

Alternative Pattern 2 – Factor times parameter (parameterized BF model)

If expected incremental loss is modeled as a percent of ultimate, we can use the parameterized BF model:

$$E[IncLoss_d] = f(d)h(w)$$

We need to solve for the $f(d)$ and $h(w)$ factors iteratively. Note that the final $f(d)$ factors won't necessarily sum to 1. Likewise, the $h(w)$ factors aren't equivalent to estimated ultimate loss. Instead, $f(d)$ and $h(w)$ are used together to estimate future incremental losses.

Compare the parameterized BF model to the chain ladder model by doing a goodness-of-fit test. If the parameterized BF model has a lower test statistic, then the loss emergence in the loss triangle is better described by the parameterized BF model.

The parameterized BF model has twice as many parameters ($2n-2$) as the chain ladder model. So, it's often over-parameterized, resulting in a higher adjusted SSE. We can improve the performance by reducing the number of parameters (which will increase SSE, but hopefully decrease Adjusted SSE).

The parameterized Cape Cod method uses one h parameter for all accident years. If the ultimate loss for each accident year is roughly the same, then this model might be an improvement over the fully parameterized BF.

This is a bit different than the normal Cape Cod model that uses the same expected loss ratio for all accident years. A possible improvement is to use a loss ratio triangle with the Cape Cod method, because a constant ultimate loss ratio over all accident years is more reasonable.

Disadvantages of the Cape Cod method:

- Requires relatively stable level of loss exposure over accident years
- Need to adjust for known exposure and price level differences between accident years

Testable Implication 3 – Linearity of the model: Residuals vs. Loss_k

- **Tests Assumption 1:** Linear loss emergence proportional to loss-to-date

Assumption 1 assumes incremental loss emergence is linearly proportional to loss-to-date.

If residuals show a non-linear pattern (e.g. positive – negative – positive), then incremental loss emergence is a non-linear function of loss-to-date.

Testable Implication 4 – Stability of development factors: Residuals vs. time

- **Tests Assumption 1:** Linear loss emergence proportional to loss-to-date

Assumption 1 uses the same development factor for all accident years, so we would expect that the appropriate development factor is stable over time.

A pattern of high/low residuals when plotted against time indicates instability and is evidence against using the same development factor for all AYs.

If the age-to-age factors appear stable, we should use all accident years when calculating the LDFs (to minimize variance). But, if it looks like the average level of the age-to-age factors is shifting over time, we could use a weighted average of the LDFs over the last n -years (e.g. 5-year weighted average LDFs).

Testable Implication 5 – Correlation of development factors

- **Tests Assumption 2:** Independence between Accident Years *

If there is significant correlation between development factors at different ages, then this is evidence against the chain ladder method.

***Conflict with Mack (1994):**

There's some overlap between Mack assumptions 1 and 2 as they're discussed in the two papers:

- In Mack (1994) (pg. 109), Mack states that the linearity assumption (assumption 1) implies that "subsequent development factors... are uncorrelated."
- In Venter Factors (pg. 832-833), Venter says that the correlation test is to check for dependencies in the triangle, which is a test on the independence assumption (assumption 2).

Just note that there's some disagreement between Venter and Mack regarding the assumption tested by the correlation of development columns.

Testable Implication 6 – Significantly high or low diagonals (calendar year effects)

- **Tests Assumption 2:** Independence between Accident Years

Run a regression of incremental losses against cumulative losses at the prior development periods and include a dummy variable for each diagonal.

If any of the dummy variables are statistically significant, then this indicates calendar year effects.

Recipes for Calculation Problems

- Correlation of Development Factors
- Parameterized BF: $f(d)h(w)$ – Constant Variance
- Parameterized BF: $f(d)h(w)$ – $\text{Var} \propto f(d)h(w)$
- Testing for Significantly High/Low Diagonals
- Goodness of Fit Test

Overview

The goal of the bootstrap model is to create a full distribution of possible loss reserve outcomes, not just a point estimate or the mean and variance like in the Mack – Chain Ladder approach. It does this by sampling residuals from the original loss triangle to create simulated loss triangles, one for each iteration in the simulation. Then, an estimated loss reserve is calculated for each iteration.

GLM Bootstrap Model

The recipes in the Exam 7 Cookbook show the different types of problems you might be asked from the bootstrap model. Below is a high-level overview of how the basic GLM bootstrap model works.

GLM Setup

- 1) Shapland uses a GLM with a log-link function and over-dispersed Poisson error distribution.

Expected Incremental Losses

$$\begin{aligned} E[q(w, d)] &= m_{w,d} \\ \ln(m_{w,d}) &= \eta_{w,d} \\ \eta_{w,d} &= \alpha_w + \sum_{d=2} \beta_d \end{aligned}$$

Variance of Incremental Losses

$$\text{Var}[q(w, d)] = \phi \cdot m_{w,d}^z$$

- 2) Next, set up the design matrix and the parameters for the model. The basic ODP bootstrap has a parameter for each accident year and development period. The GLM framework is flexible, so we can group some of those parameters together or add parameters for diagonal effects.

Y – Vector of the log-link triangle

X – Design Matrix to solve the specified model

A – Vector of the α and β model parameters

$$Y = X \times A$$

$$\text{Solution Matrix} = X \times A$$

$$\begin{bmatrix} \ln[q(1,1)] \\ \ln[q(2,1)] \\ \ln[q(3,1)] \\ \ln[q(1,2)] \\ \ln[q(2,2)] \\ \ln[q(1,3)] \end{bmatrix} = \begin{bmatrix} 1 & - & - & - & - \\ - & 1 & - & - & - \\ - & - & 1 & - & - \\ 1 & - & - & 1 & - \\ - & 1 & - & 1 & - \\ 1 & - & - & 1 & 1 \end{bmatrix} \times \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \beta_2 \\ \beta_3 \end{bmatrix}$$

$$\begin{bmatrix} \ln[m_{1,1}] \\ \ln[m_{2,1}] \\ \ln[m_{3,1}] \\ \ln[m_{1,2}] \\ \ln[m_{2,2}] \\ \ln[m_{1,3}] \end{bmatrix} = \begin{bmatrix} 1 & - & - & - & - \\ - & 1 & - & - & - \\ - & - & 1 & - & - \\ 1 & - & - & 1 & - \\ - & 1 & - & 1 & - \\ 1 & - & - & 1 & 1 \end{bmatrix} \times \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \beta_2 \\ \beta_3 \end{bmatrix}$$

- Once the model is all set up, fit the model to the triangle of incremental losses using iterated least squares or maximum likelihood. This gives us the best-fitting parameters.
- Using the best-fitting parameters, calculate the fitted incremental loss triangle.

$$\ln(m_{w,d}) = \eta_{w,d} = \alpha_w + \sum_{d=2} \beta_d \quad m_{w,d} = e^{\eta_{w,d}}$$

- Now, calculate the unscaled Pearson residuals between the original triangle of actual incremental losses and the triangle of fitted incremental losses.

$$r_{w,d} = \frac{q(w,d) - m_{w,d}}{\sqrt{m_{w,d}^z}}$$

- Use the unscaled Pearson residuals to calculate the scale parameter.

$$\phi = \frac{\sum r_{w,d}^2}{N - p} \quad \begin{array}{l} N \rightarrow \# \text{ incremental values} \\ p \rightarrow \# \text{ parameters } (\alpha\text{'s, } \beta\text{'s, hetero-adj parameters}) \end{array}$$

- Calculate the standardized Pearson residuals by applying the Hat Matrix adjustment factor to the unscaled Pearson residuals.

$$r_{w,d}^H = r_{w,d} \cdot f_{w,d}^H \quad \text{where } f_{w,d}^H = \sqrt{\frac{1}{1 - H_{w,d}}}$$

Running the Bootstrap Model

- For each iteration, sample with replacement from the standardized Pearson residuals and calculate the sampled loss triangle.

$$q^*(w,d) = r^* \cdot \sqrt{m_{w,d}^z} + m_{w,d}$$

- For each iteration, refit the GLM to the incremental losses to get new parameters, specific to this iteration's sampled loss triangle.
- Use the new parameters to calculate the projected unpaid incremental losses.

$$\ln(m_{w,d}^{iter}) = \alpha_w^{iter} + \sum_{d=2} \beta_d^{iter}$$

- Add process variance to the projected unpaid incremental losses with the Gamma distribution.

$$q^{sim}(w,d) \sim \text{Gamma}(\text{mean} = m_{w,d}^{iter}, \text{var} = \phi \cdot m_{w,d}^{iter})$$

Shapland Bootstrap Excel Files

I highly recommend that you spend some time to see how the Excel files referenced in the syllabus that accompany the Shapland paper work. It will give you a much clearer idea of how the bootstrap model works. I particularly recommend looking at the "GLM Framework 6.xlsm" file.

Advantages of the GLM Framework

- Can tailor the model to the statistical features of the data (e.g. can add a diagonal parameter, γ_k)
- Can use fewer parameters to avoid over-parameterization (e.g. fewer AY parameters)

- Can model data that's not in a loss triangle (e.g. missing data for the first few diagonals)

Disadvantages of the GLM Framework

- Simulation is slower because the GLM must be solved for in each iteration
- Can't directly explain the model using LDFs

ODP Bootstrap Model (Simplified GLM)

The simplified GLM approach can be used if the GLM bootstrap model has a parameter for each accident year and development period and uses an over-dispersed Poisson error distribution.

Instead of using the GLM to calculate the parameters, use the all-years volume-weighted LDFs to calculate expected incremental losses, $m_{w,d}$. After creating the sample loss triangles using resampled residuals, calculate new LDFs and use them to calculate the projected unpaid incremental losses. Process variance is incorporated the same way with the Gamma distribution.

Advantages of the ODP Bootstrap (Simplified GLM)

- Can use the simpler LDF method and the model will still be based on the GLM framework
- Using LDFs makes the model more easily explainable to others
- The GLM uses a log-link and may not work with negative incrementals, but the simplified GLM will still get a solution

Disadvantages of the ODP Bootstrap (Simplified GLM)

- Unable to adjust for calendar-year effects
- Requires many parameters and can over-fit the data

Practical Issues

Negative Incremental Values

Because a log-link is used, the GLM framework requires incremental losses to be greater than zero.

Adjustment 1: Sum of column of incremental losses is positive

- Calculate the log-link triangle but adjust the calculation for negative incremental losses to $-\ln[abs(IncLoss)]$

Adjustment 2: Sum of column of incremental losses is negative

- Shift all the incremental losses up by the size of the largest negative incremental
- Run the model
- Shift all the fitted incremental losses back down

Negative values during simulation of process variance:

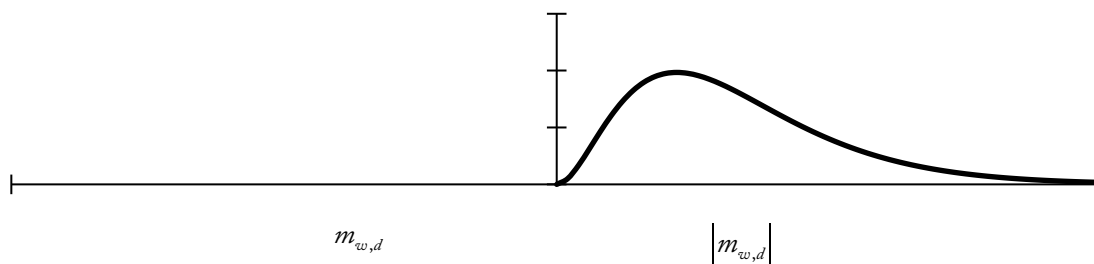
An issue that can arise is if the model simulates negative incremental losses, $m_{w,d}$. This is a problem because the process variance step can't handle negative incremental losses because the variance of the Gamma distribution would be negative.

Shapland shows two possible ways to adjust the model to handle negative $m_{w,d}$ values during simulation on pg. 21-22. For both options, a value is simulated from a modified version of the Gamma distribution, using the absolute value of the negative expected loss $m_{w,d}$. Both options transform the distribution so that the simulated value has an expected value equal to the negative incremental loss $m_{w,d}$.

Below is a graphical view of how these two adjustments work:

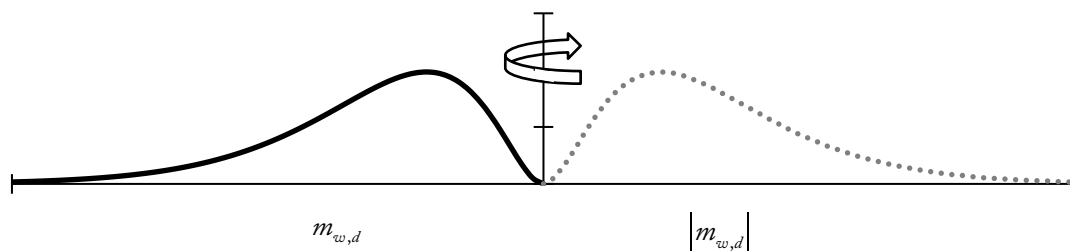
Distribution based on the absolute value of $m_{w,d}$

$$\text{Gamma}(\text{mean} = |m_{w,d}|, \text{var} = \phi \cdot |m_{w,d}|)$$



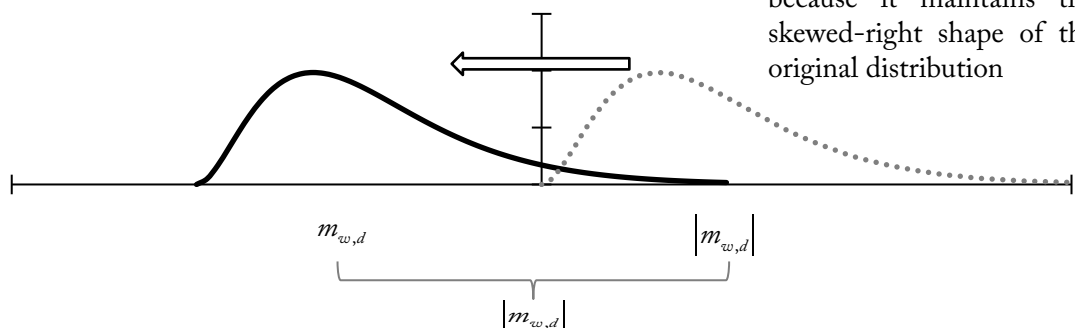
Option 1: Change the sign of the simulated value

$$-\text{Gamma}(\text{mean} = |m_{w,d}|, \text{var} = \phi \cdot |m_{w,d}|)$$



Option 2: Shift the entire distribution to have a mean of $m_{w,d}$

$$\text{Gamma}(\text{mean} = |m_{w,d}|, \text{var} = \phi \cdot |m_{w,d}|) + 2m_{w,d}$$



Note: Option 2 is preferred because it maintains the skewed-right shape of the original distribution

Options to handle extreme outcomes caused by negative incremental losses:

1. Identify and remove extreme iterations from results
2. Recalibrate the model to correct the source of negative incremental losses (e.g. model salvage and subrogation separately)
3. Limit incremental losses to a minimum of zero

Non-Zero Sum of Residuals

If we desire an average residual of zero, then we can shift all the residuals up (or down) by a constant so that the sum of the shifted residuals is zero. Then, sample from the shifted residuals.

Using N-Year Weighted Average

GLM Bootstrap Model:

- Only use the most recent $N+1$ diagonals to parameterize the model

ODP Bootstrap Model:

- Calculate N -year LDFs and use to calculate residuals for the most recent $N+1$ diagonals
- Resample residuals for the *whole* triangle (because we need to calculate cumulative losses)
- For each sample triangle, calculate new N -year LDFs and use to project future incremental losses

Missing Values

GLM Bootstrap Model:

- Missing data simply reduces the number of observations in the data.

ODP Bootstrap Model:

- Estimate the missing value from surrounding values, OR
- Modify LDFs to exclude missing values

Outliers in the Original Data

- Can remove the value and treat like a missing value, OR
- Identify and exclude outliers from the LDF calculation and residual calculations, but re-sample the corresponding incremental loss for the sampled triangles.

Heteroscedasticity

Stratified sampling:

Only sample residuals from within the heteroscedasticity group.

Disadvantage - Some groups may have few residuals. This limits the variability possible in the model.

Hetero-adjustment factors based on standard deviation of residuals:

The hetero-adjustment factor adjusts the standardized Pearson residuals to the average residual standard deviation for the triangle as a whole. With all the residuals at the same level of variability, the bootstrap model can sample residuals from the *entire* triangle.

Hetero-adjustment factors based on different scale parameters:

Instead of using standard deviations of residuals, use the ratio of the square roots of the overall scale parameter and the scale parameter for the groups based on age. Then, the hetero-adjustment factors are used the same way as above during simulation.

Heteroecthesious Data**Partial first development period data:**

- Reduce future incremental losses for the latest accident year to correspond to the earned exposure.

Partial last calendar period data:

- a) Annualize exposures in the last diagonal so they're consistent with the rest of the triangle.
- b) Calculate the fitted triangle and residuals.
- c) During the ODP bootstrap simulation, calculate and interpolate LDFs from the fully annualized sample triangles.
- d) The last diagonal of the sample triangle should be adjusted to de-annualize incrementals on the latest diagonal.
- e) Project future values by multiplying the interpolated LDFs with the new cumulative values.
- f) Reduce the future incremental values for the latest accident year to remove future exposure (because this is a partial first development period).

Exposure Adjustment

If there is rapid growth or a book is put in runoff, the exposure level can change significantly year-to-year.

- a) Divide all loss data by exposures for each accident year to get pure premium.
- b) Run the model based on pure premium.
- c) At the end, multiply results by the exposure level to get the total value.

Advantage – In the GLM framework, with the exposure adjustment, we can use fewer AY parameters.

Tail Factors

Instead of using a deterministic tail factor, we can assign a distribution to the tail factor parameter. This makes the tail factor stochastic within the bootstrap model.

Parametric Bootstrap

If there is a lack of extreme residuals, we can fit a distribution to the residuals. Instead of sampling with replacement directly from the residuals, we sample residuals from the parameterized residual distribution. This will allow for more extreme residuals in the sampled triangles.

Diagnostics

You should understand the different diagnostics used to assess the model reasonableness. You should be able to read graphs and output and draw conclusions about the model. Make sure you know what to expect from each diagnostic. Definitely review the graphs and tables in the Shapland appendices.

Purpose of diagnostics:

- Test model assumptions
- Gauge how well the model fits the data
- Help guide adjustments of model parameters

Residual Graphs

Residuals should be random around zero. You should look at the following types of residual graphs:

- Residuals vs. Calendar Year
- Residuals vs. Development Period (look for heteroscedasticity)
- Residuals vs. Accident Year
- Residuals vs. Fitted Incremental Loss

Normality Test

This test compares the residuals to the normal distribution. Normality isn't required, but Shapland does mention this as a diagnostic to review. If residuals are close to normal, you should see:

- Normality plot with residuals in line with the diagonal line (normally distributed)
- High R^2 value and p-value greater than 5%

AIC and BIC can be used to compare different models to see which model is closest to Normal while penalizing for extra parameters (a lower value is better):

$$AIC = 2p + n \left[\ln \left(\frac{2\pi \times RSS}{n} \right) + 1 \right] \qquad BIC = n \times \ln \left(\frac{RSS}{n} \right) + p \times \ln(n)$$

Outliers

Use a box-whisker plot to help identify outliers. It's important to be careful about removing outliers. If outliers represent scenarios that can't be expected to ever happen again, then it may make sense to remove them. But, "outliers" may simply be extreme, but realistic, values. If this is the case, then the "outlier" should stay in the data.

Parameter Adjustment

The GLM Bootstrap model is more flexible, so we can look at the residual graphs and adjust the parameters in the model by adding or removing additional parameters. For instance, if there is a trend in the average residuals by calendar year, then we might add a calendar year parameter to the model.

Model Results

You should see the following trends in the estimated unpaid model results by accident year:

Standard Error:

- Standard error should increase from older to more recent years
- Total standard error should be larger than the standard error for any individual year

Coefficient of Variation:

- CoV should decrease from older to more recent years
- Total CoV should be less than any individual year
- CoV could rise in the most recent years due to:
 - Parameter uncertainty in the most recent years could overpower process uncertainty
 - The model could be overestimating uncertainty (might use the BF method instead)

Multiple Models

Model risk is the risk that the model doesn't reflect the "true" loss-generating process for future losses. Weighing together multiple reasonable models helps address model risk.

Correlation

Location Mapping

For each iteration:

- a) Sample residuals for segment 1
- b) Track the location in the residual triangle where each sampled residual was taken
- c) For all other segments, sample the residuals from their residual triangles using the same locations

Advantages:

- The method is easily implemented
- Doesn't require an estimated correlation matrix

Disadvantages:

- Requires all business segments to have the same size data triangles with no missing data
- Correlation of original residuals is used, can't test other correlation assumptions

Re-sorting

Uses algorithms such as a copula to add correlation.

Advantages:

- Data triangles can be different shapes/sizes by segment

- Can use different correlation assumptions
- Different correlation assumptions may have other beneficial impacts on the aggregate distribution
 - Ex. Can use a copula with a heavy tail distribution to strengthen the correlation between segments in the tails, which is important for risk-based capital modeling

Recipes for Calculation Problems

- Over-Dispersed Poisson GLM Setup
- Expected Incremental Losses from GLM
- Standardized Pearson Residuals
- ODP Bootstrap Model
- Negative Incremental Values
- Heteroscedasticity Fix: Standard Deviation
- Heteroscedasticity Fix: Scale Parameter
- Multiple Models: Same Random Variables
- Multiple Models: Independent Variables

A Model for Reserving Workers Compensation High Deductibles

Overview

The typical loss reserve methods are appropriate for ground-up losses. Siewert focuses on different methods that can be used to estimate unpaid loss reserves for different layers of losses, such as limited losses or losses above a high deductible. One of the main challenges Siewert addresses is how to calculate LDFs that are consistent between unlimited, limited and excess layers.

Excess Loss Reserving Methods

Loss Ratio Method

Siewert applies the loss ratio method by account to reflect differences in account characteristics. There are two components to the expected excess losses:

- **Per Occurrence Charge** – Expected losses above the per-occurrence deductible
- **Aggregate Loss Charge** – Expected losses in the deductible layer (below the per-occurrence limit) that are above the aggregate limit

Advantages:

- Useful for immature accident years when loss data is thin
- Loss ratio estimates can be consistent with pricing

Disadvantages:

- Ignores actual loss experience, not that useful after several years development
- May not reflect account characteristics properly if development emerges differently due to written exposures

Implied Development

The ultimate excess loss estimate is the difference between the unlimited and limited ultimate loss estimates (calculated using the unlimited and limited losses and LDFs).

$$UL_t^{excess} = UL_t^{unlimited} - UL_t^{limited}$$

The limited LDFs used to calculate $UL_t^{limited}$ need to reflect indexed limits adjusted for inflation over time. This is because using a constant deductible for all accident years implies increasing excess losses.

Advantages:

- Can estimate excess loss for early maturities, even if no excess losses have emerged yet
- Limited LDFs are more stable than excess LDFs used for direct development

Disadvantages:

- Misplaced focus – we would like to explicitly recognize excess loss development

Direct Development

Apply the XSLDF to excess loss to estimate ultimate excess loss. The XSLDF should be consistent with the limited and unlimited LDFs.

Advantages:

- Focuses explicitly on excess loss development

Disadvantages:

- XSLDFs are often highly leveraged and volatile, especially for immature periods
- Direct development doesn't produce an estimate if no excess loss has emerged yet

Credibility Weighting (Bornhuetter-Ferguson)

Excess ultimate loss estimate is a credibility weighting of the actual experience and the expected losses (loss ratio method).

Advantages:

- Can tie to the pricing estimate for immature periods when losses haven't emerged
- More stable estimate over time

Disadvantages:

- Ignores actual loss experience to the extent of the complement of credibility

Development Model

The first way we can get LDFs for different limits is to calculate them directly using full coverage loss experience. This way, we can directly create loss triangles with losses limited at different limits.

Indexed Limits

Before calculating the limited LDFs from a limited loss triangle, we need to index the limits for prior accident years. This is important so that the proportion of limited-to-excess loss around the limit is consistent from year to year.

To do this, Siewert fits an exponential curve to loss severity by accident year to calculate the annual severity trend. Then, he uses the selected severity trend to trend the loss limit backwards. Once we have the indexed limits, we can create a loss triangle with losses limited at the appropriate indexed limits and calculate the limited LDFs.

See the example below about how to calculate indexed limits:

Example 1

Given:

- Annual loss severity trend is 10%
- Deductible for an account is \$250,000

Calculate the indexed limits to use for a five-year limited loss triangle.

Solution

First, set the limit for the most recent accident year to the stated limit, \$250k. Then, trend the limits backward for each prior accident year.

Accident Year	Indexed Limit
2012	170,753
2013	187,828
2014	206,612
2015	227,272 ← $= \frac{250,000}{1.10}$
2016	250,000

Tail Factors

Siewert uses the inverse power curve to get consistent tail factors by limit. First, he fits an inverse power curve to the unlimited LDFs and select an appropriate stopping point beyond which there's no further development (e.g. 40 years). He selects the stopping point so that the inverse power curve tail factor is consistent with a tail factor based on an extended loss triangle.

To get the tail factors for the other limits, he fits the inverse power curve to the limited LDFs at each limit up to the stopping point and compound the age-to-age factors from the fitted curves.

Relationships between unlimited/limited/excess LDFs and severity relativities

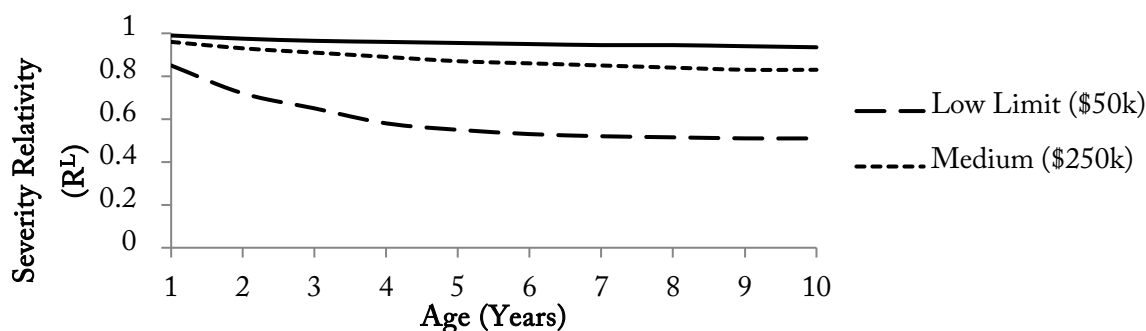
There is a relationship between limited/excess LDFs and unlimited LDFs using loss severity relativities:

$$R_t^L = \frac{\text{Severity Limited to limit } L \text{ at age } t}{\text{Unlimited Severity at age } t}$$

Severity relativities should have the following relationships:

- **Severity relativity should decrease as age increases**
 - This is because more losses are capped at the per-occurrence limit as age increases
- **Severity relativity should be higher for a larger limit**
 - This is because a higher limit means less of the loss is capped, so the relativity is higher

You can see these relationships in the following graph of severity relativities by age and limit:



Below are the key relationships:

$$XSLDF_{t_1-t_2}^L = LDF_{t_1-t_2} \cdot \frac{(1 - R_{t_2}^L)}{(1 - R_{t_1}^L)}$$

$$LDF_{t_1-t_2}^L = LDF_{t_1-t_2} \cdot \frac{R_{t_2}^L}{R_{t_1}^L}$$

The unlimited LDF is a weighted average of the limited and excess LDFs. With this relationship, we can split the unlimited loss development consistently between development below and excess the limit.

$$LDF_{t_1-t_2} = R_{t_1}^L \cdot LDF_{t_1-t_2}^L + (1 - R_{t_1}^L) \cdot XSLDF_{t_1-t_2}^L$$

By using the severity relativities alone to adjust the LDF, we implicitly assume no further claim count development.

Distribution Model

If we use the loss data directly, we'll find instances where limited development sometimes is greater than the unlimited development. This will happen if the limited severity relativity increases from one age to the next (remember, the severity relativity should decrease as age increases).

Instead of using the data directly, we can fit a separate severity distribution to losses at each age (the severity distribution for losses at 12 months will be different than the ultimate severity distribution). Then, we can calculate the severity relativities at each age for any limit we're interested in.

Siewert calculates the unlimited LDFs directly from the fitted severity relativities. Then, using the fitted severity relativities and these unlimited LDFs, we can then calculate consistent LDFs at any other limit.

Fitting the Model

Siewert uses a Weibull distribution and estimates the parameter by minimizing the chi-square error between the actual and expected severity relativities. Also, Siewert constrains the model so that the unlimited expected severity is equal to the actual unlimited severity at that development age.

Claim Count Development Assumption

In the development model, we started with selected unlimited LDFs based on unlimited loss data. A key difference here is that we're calculating the LDFs directly from the modeled severities. By using the severity relativities alone to calculate the LDFs, we implicitly assume no further claim count development. Because of this, an important assumption is that there's no further claim count development.

In Siewert (see pg. 232), he discusses how we can use the modeled severities to calculate the LDFs, but specifies that if there is claim count development for the earlier maturities, it needs to be reflected.

Partitioning Loss Development above and below the deductible

Because the unlimited LDF is a credibility weighting of the limited and excess LDFs, we can partition expected future development between development above and below the deductible.

As age increases, a larger portion of the expected development will be due to excess development above the limit.

$$\begin{aligned}\%Unpaid &= 1 - \frac{1}{LDF} = \frac{LDF - 1}{LDF} \\ &= \frac{R_t^L \cdot (LDF^L - 1) + (1 - R_t^L) \cdot (XSLDF^L - 1)}{LDF} = \underbrace{\frac{R_t^L \cdot (LDF^L - 1)}{LDF}}_{\text{Development below the deductible}} + \underbrace{\frac{(1 - R_t^L) \cdot (XSLDF^L - 1)}{LDF}}_{\text{Development above the deductible}}\end{aligned}$$

Example 2

At 48 months development, below are the paid LDFs around the 250k deductible:

Unlimited LDF	Limited LDF	XSLDF
2.152	1.959	3.148

- The 250k severity relativity at 48 months is 0.838

Partition the expected future paid development above and below the 250k deductible.

Solution

$$\begin{aligned}\%Unpaid &= 1 - \frac{1}{2.152} = 53.5\% \\ &= \frac{.838 \cdot (1.959 - 1)}{2.152} + \frac{(1 - .838) \cdot (3.148 - 1)}{2.152} = 37.3\% + 16.2\%\end{aligned}$$

The expected development of 53.5% is made up of 37.3% of development below the deductible and 16.2% of development above the deductible. Put another way, 69.7% (= .373/.535) of expected future development is below the deductible and 30.3% is above the deductible.

Aggregate Excess of Loss Estimate

An aggregate limit caps losses in the deductible layer that the insured is responsible for. We can't calculate development factors for losses excess the aggregate limit because the data needed to do this is thin and not likely to be credible.

Instead, Siewert recommends using a collective risk model or Table M to estimate liabilities under an aggregate limit.

Collective Risk Model

Siewert uses the fitted Weibull model for the severity distribution and a Poisson claim count distribution for frequency in the collective risk model. The collective risk model provides expected aggregate excess loss in the deductible layer at each age and at ultimate (using the different fitted Weibull models for each age). Then, the expected excess losses are used to calculate the aggregate excess of loss LDFs.

Because losses excess the aggregate are so volatile, Siewert recommends using the BF method.

These are the key relationships you should notice:

- **Aggregate excess loss development drops off faster for smaller per-occurrence deductible limits**
 - Because later development is more likely for larger claims that already are over the deductible (so they don't impact the aggregate limit)
- **Higher aggregate limits have more leveraged LDFs**
 - Because there are fewer losses excess the aggregate limit when the aggregate limit is high, especially at earlier periods

Table M Approach

An aggregate limit is similar to a maximum premium limit used in retrospective rating plans with Table M. This is covered in Exam 8, so I wouldn't expect a significant question on the exam about this approach, but Siewert does show an example calculation in the appendix.

An advantage of this approach compared to the collective risk model approach is that it can be more practical.

Service Revenue

In a high deductible program, the insured is responsible for the deductible losses that are below the aggregate limit, but the insurer still services those claims.

- **Service Revenue** - Reflects the costs the insurer charges the insured for servicing those claims.
- **Service Revenue Asset** - The expected ultimate service revenue minus the recoveries for service revenue to-date.

Recipes for Calculation Problems

- Loss Ratio Method
- Implied Development Method
- Direct Development Method
- Credibility Weighted Method
- Unlimited/Limited/Excess LDF Relationships
- Distributional Model
- Aggregate Excess of Loss Ultimate
- Aggregate Excess of Loss Ultimate: Table M
- Service Revenue Asset

Overview

The goal of this paper is to show the relationship between **loss trend**, a **claim size (severity) model** and **cumulative loss development factors**. With this relationship, we can adjust CDFs from a “base” layer to another layer and from one cost level to another.

Trend

The loss trend factors we use are for the ground-up unlimited loss layer and apply to cumulative loss, not incremental loss. We create a trend table of the cost level indices. This way, we can reflect trends that act in different directions such as an accident year trend or a calendar year trend.

Claim Size (Severity) Model

The loss severity distribution is used to calculate limited expected values (LEVs) at different limits, which are used to adjust CDFs from one layer to another.

We start with a severity distribution at the current cost level for each development period with different parameters for each age. This is similar to how Siewert uses a different Weibull severity distribution for each age. Then, using the trend, we can find the appropriate parameters (unlimited expected claim size) for the severity distributions at the prior accident year cost levels by trending backwards.

Loss Development

Loss development factors are calculated as volume-weighted LDFs and CDFs using the chain ladder method.

Relationship Between Layers

Remember, the goal of this paper is to use the relationship between trend, a claim size model and loss development to calculate appropriate CDFs for any other layer and cost level.

For instance, we might have a triangle of ground-up loss data limited to \$1,000,000 and a loss triangle from a book of business with a \$500,000 high deductible. We can use the relationship between these three components with the method below to calculate the appropriate CDFs that we can apply to the latest diagonal to estimate ultimate excess loss for the book of business.

Getting CDFs at a basic limit

Normally, we calculate the LDFs directly from an unadjusted loss triangle and use them to develop losses to ultimate for all accident years. BUT, this is only appropriate if:

- The loss triangle is ground-up unlimited AND
- Trend *only* acts in the accident-year direction

Often this isn't the case, so we first adjust the loss triangle to get data to the same cost level and to a basic per-occurrence limit where we believe the data is credible for estimating LDFs.

To do this, we first calculate the unlimited expected claim size at each age trended backwards. Then, we can calculate the LEVs to adjust the loss data. We need the LEVs at the limit of the data and accident year cost level (where the loss data is now). We also need the LEVs at the basic limit and the most recent accident year cost level.

With the LEVs, we can adjust the loss data so that the whole triangle is now at the basic limit and current accident year cost level. There are **two adjustments** going on at the same time:

- **Loss Limit** - Adjust from the loss limit of the original data to the basic limit
- **Cost Level** - Adjust from the separate accident year cost levels to a common cost level

$$C'_{AY,k} = C_{AY,k} \cdot \frac{\text{LEV}[B|\theta_{n,k}]}{\text{LEV}[L|\theta_{AY,k}]} \quad \text{Adjusted Loss}_{AY,k} = \text{Loss}_{AY,k} \cdot \frac{\text{LEV}[B_{\text{Latest } AY,k}]}{\text{LEV}[L_{AY,k}]}$$

Then, we calculate the LDFs and CDFs from the adjusted loss triangle. Now we have a development pattern that's at the basic limit and a common cost level.

Getting CDFs at any other layer and accident year cost level

Once we've got the development pattern at a basic limit and common cost level, we can use the LEVs from the claim size model to adjust the development pattern to any other limit and cost level. Most importantly, we can get the CDFs that we can apply to the most recent loss diagonal to develop losses.

The formula below uses the relationship between trend, claim size model and CDFs to calculate the adjusted CDFs:

$$F_{AY,k}^X = F_{n,k}^B \cdot \frac{\text{LEV}[X|\theta_{AY,\infty}]/\text{LEV}[B|\theta_{n,\infty}]}{\text{LEV}[X|\theta_{AY,k}]/\text{LEV}[B|\theta_{n,k}]} \quad CDF^X = CDF^{\text{Basic}} \cdot \frac{\text{LEV}[X_{AY,Ultimate}]/\text{LEV}[B_{\text{Latest } AY,Ultimate}]}{\text{LEV}[X_{AY,k}]/\text{LEV}[B_{\text{Latest } AY,k}]}$$

Again, there are two adjustments going on in this formula:

- Adjust CDFs from the basic limit to layer X
- Adjust CDFs from the common cost level to separate accident year cost levels along the diagonal

Now, we can use these adjusted CDFs with a loss reserving method to estimate loss reserves for a loss triangle at the new layer, layer X.

Simplified Model

A disadvantage of this method is that it requires a claim size model at each development age. We might only know the severity distribution and the expected unlimited severity at ultimate.

In this case, we can use a simplified model (below) to adjust the basic limit CDFs to the new layer for the most recent diagonal:

$$F_{AY,k}^X = F_{n,k}^B \cdot \frac{\text{LEV}[X | \theta_{AY,\infty}]}{R_k \cdot \text{LEV}[B | \theta_{AY,\infty}]}$$

$$\text{Where: } R_k = \frac{\text{LEV}[X | \theta_{AY,k}]}{\text{LEV}[B | \theta_{AY,k}]}$$

In this model, R_j is the ratio between the limited severities at layer X and layer B. This is similar to the severity relativity in the Siewert paper, but remember Siewert always looks at the severity relativity of limited-to-unlimited loss. Here, we're looking at the severity relativity between limited losses at two different limits.

Besides using a ratio for the denominator, note that this simplified variation doesn't do the adjustment from one cost level to another. This method makes the assumption that the differences in cost level are immaterial to the ratios of the LEVs by layer.

If the new layer, X, is lower than the basic limit, then $R_j < 1$ and Sahasrabuddhe points out that one should see the following properties:

- **R_j decreases as age increases.**
 - At earlier maturities, most of the development will be in the lower layer below X, so R_j will be close to 1.
 - At later development periods, there will be more development in the layer between limits X and B. This means R_j will decrease for later development periods.
- **$R_j > U = \lim R_j$ at ultimate, calculated as the ratio of LEVs at ultimate between layer X and B.**
 - This follows from the bullet above. R_j monotonically decreases as maturity increases until losses are at ultimate and the ratio R_j is at its minimum, U.
 - Also, Sahasrabuddhe points out that U can be calculated as:

$$U = R_j \cdot \frac{CDF^X}{CDF^B}$$

- In this formula, the ratio of CDFs develops both the numerator and denominator of R_j to ultimate, giving the ultimate ratio U.
- **If the base layer CDFs are ground-up unlimited, then the upper limit of R_j is $\max(R_j) = U \times CDF$.**
 - In this case, the Basic Limit, B, is Ground-Up Unlimited.
 - For any maturity j before ultimate, the maximum of what the ratio R_j can be is when losses limited at X are fully developed as of time j , and all future development after j is in the layer excess limit X. This would mean $CDF_{\text{Layer X}} = 1.00$.
 - In this case, the ratio at ultimate can be calculated as follows by plugging into the formula from above:

$$U = R_{j,\max} \cdot \frac{1.00}{CDF^B}$$

- Rearranging this, and since layer B is ground-up unlimited, we have:

$$R_{j,\max} = U \cdot CDF$$

Assumptions

- Actuary must select the basic limit
- Requires a claim size model (but with the simplified model we only need it at ultimate)
- Loss triangle must be adjusted to basic limit and a common cost level

More difficult assumptions

- Claim size models at earlier maturities aren't usually available
 - The simplified version only uses the ratio of LEVs at earlier maturities which could be easier to estimate
- Must calculate trend indices appropriate for cumulative losses, which may be difficult because:
 - Trend typically impacts incremental losses
 - Trend on reported losses and the timing of trend on case reserves is difficult to get at

Interpreting Results

We can see how this approach impacts the development pattern if we take a limited loss triangle at limit L and compare the CDFs calculated directly from the triangle to the CDFs calculated from the method above (first adjusting to a basic limit and common cost level, then adjusting back to limit L and the cost level for the latest diagonal).

Examples 1 and 2 in the paper appendix shows this comparison. Both examples use the same data but Example 2 uses a severity model with larger claim sizes. Example 2 shows greater differences between the calculated CDFs with the Sahasrabuddhe method and the unadjusted CDFs (from the original loss triangle).

Key relationships

- Differences are greater for larger expected unlimited claim size which increases the expected loss in the layer between the basic limit and the new limit.
- Differences are greater where trend and/or loss development act over longer time periods (long-tailed lines) or when the loss trend is higher.

The differences are smaller for short-tail lines, low trend rates and limits above the working layer (limits significantly higher than the average severity).

Recipes for Calculation Problems

- Determine LDFs at Basic Limits
- Calculate Layer LDFs from Basic Limit LDFs
- Calculate Layer LDFs from Basic Limit LDFs: Simplified Model

Teng and Perkins

Estimating the Premium Asset on Retrospectively Rated Policies

Overview

Teng and Perkins is a calculation-heavy paper focused on how to estimate the premium asset for retrospectively rated policies. The key to this method is the relationship between loss development and premium development (the PDL ratios).

Make sure to review the exhibits at the end of the paper thoroughly. They show all the steps for calculating the final premium asset from the original premium and loss triangles.

Retro Rated Policies

Reasons why Retro Rated Policies are Popular

- Premium is returned to the insured for good loss experience
- Cash flow advantage - The insured can pay premiums as losses are reported or paid, instead of up front
- More risk is shifted to the insured – Final premium varies with loss experience (good for insurer)

Retro Premium Formula

The following formula gives the premium for a retro policy at the n^{th} adjustment:

$$Prem_n = (Basic\ Prem + Capped\ Loss_n \cdot Loss\ Conversion\ Factor) \cdot Tax\ Multiplier$$

- Basic Premium (BP) – Covers company expenses, the insurance charge for the max/min premium, and the excess loss charge for a per-accident loss limit
- Capped Losses (CL_n) – Losses that contribute to additional premium
 - Losses are capped at a min/max corresponding to a min/max premium for the policy, and individual claims may be capped as well.
- Loss Conversion Factor (LCF) – Covers loss adjustment expenses
- Tax Multiplier (TM) – Covers state premium taxes and other state assessments

Fitzgibbon Method

The idea of the Fitzgibbon method is that the retro formula is essentially a linear equation and instead of matching the individual policy retro rating parameters to the individual loss experience, we can run a linear regression to get the average retro rating parameters for the linear equation.

The key assumption is that the premium is assumed to be a linear function of incurred losses. If you look at the retro rating formula, it's in the form of a linear equation.

The Fitzgibbon method, the way Feldblum shows it, uses this approach:

$$\boxed{Y = A + B \cdot X}$$

$$A = \frac{\text{Basic Premium}}{\text{Standard Premium}}$$

$$B = \text{Premium Responsiveness}$$

$$\underbrace{\frac{\text{Retro Premium}}{\text{Standard Premium}}}_{\text{Retrospective Premium \%}} = A + B \cdot \underbrace{\frac{\text{Ult Loss}}{\text{Standard Premium}}}_{\text{Standard Loss Ratio}}$$

Fitzgibbon estimates the parameters for the model, A and B, using a linear regression of retrospective adjustments for mature policies against the standard loss ratios for those policies.

Reasons why the slope (B) doesn't equal 1

- Losses over the loss limit or beyond the max premium don't increase the retrospective premium
- The minimum premium is greater than the basic premium for some plans
- The loss conversion factor (LCF) and tax multiplier (TM) change the slope

Disadvantages of the Fitzgibbon Method

- Fitzgibbon calculates the *ultimate* premium estimate based on cumulative loss-to-date. If the retro premium-to-date has been less responsive than expected, there's no way to get "back on track" because the method only estimates ultimate premium based on estimated ultimate losses.
- Fitzgibbon uses a constant slope (B) for premium sensitivity, but we would expect the slope to decrease at higher overall loss ratios due to loss limits and maximum retrospective premiums.

Example 1

- Losses at 36 months are 1,000
- Expected Ultimate Loss is 5,000
- $A = 1,000$
- $B = 1.2$

Calculate the future premium based on the following values of retrospective premium as of 36 months:

- $\text{Prem}_{36} = 2,200$ (Expected retro premium for $\text{Loss}_{36} = 1,000$)
- $\text{Prem}_{36} = 1,500$

Solution

$$\text{Est Ult Prem} = 1,000 + 1.2 \times 5,000 = 7,000$$

Retro Premium	Premium at 36 Months	Est. Ult Premium	Future Premium
Expected	2,200	7,000	4,800
Lower-than-Expected	1,500	7,000	5,500

Using the Fitzgibbon method, the estimated ultimate premium only depends on the overall ultimate loss estimate. If the premium responsiveness is less-than-expected for losses-to-date (the second situation), the estimated future premium will increase so that the overall estimated ultimate premium stays the same.

The PDL method fixes this problem by estimating future retro premiums directly using future expected losses.

Teng and Perkins PDL Method

The PDL method is a bit like the BF method. It focuses on future premiums based on future expected losses, independent of prior experience.

Advantages of the PDL Method

- It's directly modeled on the retrospective rating formula, so it's easily explainable
- Its focus on premium sensitivity is like how loss sensitive contracts are handled under Risk-Based Capital
- May be useful when changes to retro rating parameters distort indications from other methods

Reasons to Use the PDL Method Instead of the Chain Ladder Method

- Timeliness – We can estimate ultimate loss sooner than we can estimate retrospective premiums. Also, we can update the premium asset estimate quarterly with the PDL method.
- Retrospective premiums depend on incurred losses

Teng and Perkins PDL Assumptions

- Premium responsiveness for future adjustments is independent of the premium responsiveness of past adjustments.
- The slope of the line segment (premium responsiveness for an adjustment period) is independent of the beginning loss ratio and or the beginning retro premium ratio.

Reasons It's Difficult to Use Retro Rating Parameters from Pricing

- Different insureds have policies with different retro rating parameters (e.g., a policy for a small insured may have a lower max premium than a large insured)
- Premium taxes differ between states
- Basic premium may vary over time

General Rule for Premium Responsiveness

- Premium responsiveness decreases as a book of business matures. (larger % of losses are excluded due to loss limits and max premium)
- Premium responsiveness decreases for higher overall loss ratios. (policies are more likely to have hit max premium)

Premium Development to Loss Development (PDLD) Ratios

Method 1 – PDLD Ratios based on the Retro Formula

Advantages of the Retro Formula Method

- Responds to changes in the retro rating parameters for sold policies

Disadvantages of the Retro Formula Method

- Must select retro rating parameters, which can be difficult because parameters will vary between policies sold (Teng recommends using the average of the sold parameters)

If the retro rating parameters change significantly over time, PDLD ratios from the retro formula method should be given greater weight than those based on historical data. The retro formula can directly reflect those changes.

Method 2 – PDLD Ratios based on Historical Data

For this method, we use booked premium and reported loss development to calculate the PDLD ratios. Data should be partitioned into homogeneous groups based on account size and type of rating plan. Policies are grouped by effective date, so data is based on policy effective quarter (not accident year).

Premium for a retro-rated policy is evaluated at each “retro adjustment period.” The first retro adjustment typically uses losses at 18 months of development. Then retro adjustments occur every 12 months. Premiums are assumed to be booked 3 to 9 months after the losses used to calculate the premium for the retro adjustment.

Reviewing PDLD Ratios

- Look for trends in the historical PDLD ratios by policy quarter. An upward trend could indicate:
 - More “liberal” rating parameters that increase the capped losses and therefore increase the premium (e.g., higher maximum premium or per-accident limit).
 - Better loss experience, resulting in more losses falling within the capped limits.
- Historical PDLD ratios may be volatile after the first retro adjustment because incremental premium development may reflect incremental loss development on only a small number of policies.
- It’s possible to see negative incremental historical PDLD ratios when loss development is positive:
 - There could be upward development on a claim that already passed the per-accident limit (no change in premium) while other losses within the limits fall (causing the premium to fall).
- Compare the average of many quarters of the historical PDLD ratios to PDLD ratios from the retro formula to select a PDLD ratio.
- PDLD ratios for the first adjustment are typically greater than 1 due to the Basic Premium and the fact that few losses are capped.
- PDLD ratios should generally fall for more mature retro adjustments because more losses will be capped.

Reasons Historical PDLR Ratios may differ from the Retro Formula PDLR Ratios

- Worse (or better) than expected loss experience can cause more (or less) losses to be capped, resulting in historical PDLR ratios that are smaller (or larger) than the retro formula PDLR ratios.
- Average retro rating parameters may change over time.

Loss Capping Ratios

The loss capping ratio typically *decreases as losses mature* since more losses are outside the loss limits (aggregate limits for min/max premium and per-accident limits).

For the exam, it's important to know the difference between incremental and cumulative loss capping ratios. The retro formula uses *incremental* loss capping ratios so if the problem gives you cumulative loss capping ratios, you need to adjust them to incremental ones first. If it's unclear, make sure to state your assumption!

Cumulative Loss Capping Ratios

$$\frac{CL_n}{L_n} = \frac{\text{Capped Loss}_n}{\text{Loss}_n}$$

Incremental Loss Capping Ratios

$$\frac{\Delta CL_n}{\Delta L_n} = \frac{\text{Capped Loss}_n - \text{Capped Loss}_{n-1}}{\text{Loss}_n - \text{Loss}_{n-1}}$$

If you're given cumulative loss capping ratios, you can convert to incremental loss capping ratios. The formula is shown in section 5, which is off the syllabus, but also shows up in exhibit 5 in the appendices. I think it still could be testable and a few past CAS problems are conflicting regarding incremental vs. cumulative loss capping ratios, so it's a good idea to be prepared for it.

$$\frac{\Delta CL_n}{\Delta L_n} = \frac{\left(\frac{CL_n}{L_n}\right) \cdot \%Loss_n - \left(\frac{CL_{n-1}}{L_{n-1}}\right) \cdot \%Loss_{n-1}}{\%Loss_n - \%Loss_{n-1}}$$

Cumulative Premium Development to Loss Development (CPDLR) Ratios

Remember that CPDLR ratios relate to PDLR ratios the same way that CDFs relate to LDFs for the chain ladder method.

The goal of the CPDLR ratios is to estimate the total expected future premium development for a policy effective year by multiplying the CPDLR by the expected future loss.

Premium Asset

It's important to remember that the premium asset is calculated as the difference between ultimate premium and *current booked* premium, NOT premium booked at the prior adjustment. This is because there will be some differences between the booked premium at the evaluation date of the analysis and the premium booked at the prior adjustment period.

We should also consider the collectability of the premium asset (similar to recovery risk in Marshall) due to the possibility that the insured won't pay future premium development.

Enhanced PDL Method – PDL with a constant

CPDLD₁ in the Teng and Perkins PDL method includes both the impact of losses on the retro premium and the basic premium. One of the advantages of the Fitzgibbon method is that the basic premium portion was broken out and reflected as a fixed amount.

Feldblum proposes an enhancement to the PDL method by removing the basic premium portion from the CPDLD₁. This way, the adjusted CPDLD₁ only reflects future premium development due to future loss development and the basic premium is added afterwards as a fixed amount.

Recipes for Calculation Problems

- PDL Ratios – Retro Formula
- PDL Ratios – Empirical
- Premium Asset
- Cumulative to Incremental Loss Capping Ratios
- Fitzgibbon Method
- Premium Asset – PDL Method Enhancement

Stochastic Loss Reserving Using Bayesian MCMC Models (2nd Edition)

Overview

The main goal of this paper is to backtest different stochastic reserve models to determine which models best produce a predictive distribution of ultimate loss. The paper shows that the traditional reserve models fail to validate. Then, it builds a set of Bayesian MCMC models that work better.

Finally, it shows how to determine the total reserve liability (*Best Estimate + Risk Margin*) using a given stochastic model.

The final model (IPI) jointly fits to both paid and incurred loss triangles to significantly reduce uncertainty in estimated ultimate loss. The lower uncertainty reduces the risk margin.

Retrospective Analysis

The goal is to evaluate a model's **reputation** for producing predictive ultimate loss distributions that are well-calibrated to the true ultimate loss distributions across a dataset of completed loss triangles.

Below are the key steps for a retrospective analysis of a reserve model:

1. For each loss triangle dataset:
 - a. Fit the model to the upper triangle of losses
 - b. Create the **predictive distribution** of ultimate loss based on the model
 - c. Calculate the **predicted percentile of outcome**, the actual ultimate loss within the predictive distribution of ultimate loss
2. Analyze the percentiles over all loss triangle datasets:
 - a. A validated model with a good reputation will show uniform predicted percentiles

p-p Plot & Histogram

If the model validates and the percentiles are uniformly distributed, we should see:

- **Histogram** - Bars of equal height
- **p-p Plot** – Sorted predicted percentiles of outcome follow expected percentiles along a 45° line

Kolmogorov-Smirnov (K-S Test)

The **K-S statistic** compares the maximum distance between the predicted and expected percentiles against a threshold. If the K-S statistic exceeds the threshold, the test fails, and we reject the null hypothesis that the percentiles are uniformly distributed.

ELPD_{Test} Statistic

elpd is a generalization of AIC for Bayesian MCMC models to evaluate model fit. It measures how probable the observed data is, given the model's posterior, and is used to compare models.

The elpd_{test} statistic is a *retrospective* measure of how probable the *lower (holdout) triangle* data is, given the model's posterior. A higher value indicates that the model predicts actual outcomes better.

Prospective Analysis and Diagnostics

The goal is to analyze how well a selected model fits a single loss triangle for a reserve analysis where the ultimate loss is unknown.

Residuals

Use standardized residuals to gauge the fit between the actual log loss and predicted log loss in the triangle:

$$r_{w,d}^j = \frac{\log(C_{w,d}) - \mu_{w,d}^j}{\sigma_d^j} \quad \text{residual}_{AY,dev}^{sim} = \frac{\log(Loss_{AY,dev}) - \mu_{AY,dev}^{sim}}{\sigma_{dev}^{sim}}$$

Box Plot

A box-and-whisker plot is a graphical view of standardized residuals by accident year or development period. For a good model fit, the interquartile range should contain zero and fall close to the standard normal interquartile range.

ELPD_{loo} Statistic

The elpd_{loo} statistic is a *prospective* measure of how probable the observed *upper-triangle* data is, given the model's posterior using a leave-one-out approach. A higher elpd_{loo} (and lower LOOIC) indicates a better model fit compared to other models on the same data.

Traditional Reserve Models

Meyers backtests the traditional Mack Chain Ladder and Bootstrap ODP (Shapland) models with the retrospective analysis. Both models have a *poor reputation* and *do not validate*.

Mack Model on Incurred and Paid

The Mack Chain Ladder model is a stochastic version of the chain ladder method based on the three Mack assumptions. It fits a lognormal distribution to the mean and standard deviation of estimated ultimate loss and uses it to calculate the predicted percentile of outcome for each loss dataset.

- **Mack Model (Paid)** – Biased high (*overestimates* the ultimate losses)
- **Mack Model (Incurred)** – Light tailed (*understates the variability* of ultimate losses)

Bootstrap ODP on Paid

We use the same process as in Shapland to fit a GLM to the paid-loss triangle and to run the ODP bootstrap to create the predictive distribution.

$$E[\tilde{I}_{w,d}] = a_w \beta_d$$

$$Var[\tilde{I}_{w,d}] = \phi a_w \beta_d$$

- **Bootstrap ODP (Paid)** – Biased high

Bayesian MCMC Models

Traditional models don't validate, so Meyers tries different Bayesian MCMC models to find ones that do.

How to Create the Predictive Distribution of Ultimate Loss

1. **Specify the model** - Set prior distributions for all initial parameters and specify the formulas for the transformed parameters ($\mu_{w,d}$ and σ_d) used for a lognormal distribution of losses ($C_{w,d}$).
2. **Create the posterior distribution** – Run the Bayesian MCMC software on the loss triangle with the model specification to create a posterior distribution for all parameters using Bayes theorem.
3. **Sample from the posterior distribution** - Draw 10,000 parameter sets $\{\alpha_1, \alpha_2, \dots, \log elr, \beta_1, \beta_2, \dots\}$ from the posterior distribution and then calculate the transformed parameters $\{\mu_{w,d}$ and $\sigma_d\}$
4. **Simulate ultimate losses** - For each sampled parameter set, simulate the ultimate loss for each accident year and sum over all accident years to calculate the total simulated ultimate loss.
5. **Create the predictive distribution of ultimate loss** - The 10,000 simulated outcomes result in an empirical predictive distribution of ultimate losses and reflect both parameter and process variance.

Cross Classified Model (CRC)

The CRC model is the base MCMC model. It has one parameter per accident year (α_w), one per development period (β_d), and *logelr*, the log of the expected loss ratio. Log premium is the exposure base.

It assumes:

- The claims settlement pattern is the same for each accident year (relaxed with the CSR model)
- Losses are independent between accident years (relaxed with the CAY model)

$$\mu_{w,d} = \log(Premium_w) + \log elr + \alpha_w + \beta_d$$

$$C_{w,d} \sim \text{lognormal}(\mu_{w,d}, \sigma_d)$$

- **Paid CRC** – Biased high
- **Incurred CRC** – Light tailed

Stochastic Cape Cod Model (SCC)

The SCC model is a stochastic implementation of the Cape Cod method using a single expected loss ratio for all accident years, so there is no α_w parameter. This is the worst model because it doesn't allow expected loss ratios to vary by year.

$$\mu_{w,d} = \log(Premium_w) + \log elr + \beta_d$$

- **Paid SCC** – Light tailed (very poor)
- **Incurred SCC** – Light tailed (very poor)

Changing Settlement Rate Model (CSR)

Key improvement over CRC:

The CSR model's γ parameter adjusts the model for changes in settlement rates across accident years (w):

- $\gamma > 0$ indicates a speedup in settlement rates over time
- $\gamma < 0$ indicates a slowdown in settlement rates over time

$$\mu_{w,d} = \log(\text{Premium}_w) + \text{logelr} + \alpha_w + \beta_d \times (1 - \gamma)^{w-1}$$

- **Paid CSR** – Corrects the CRC overstatement and better estimates the ultimate losses

Correlated Accident Year Model (CAY)

Key improvement over CRC:

The CAY model's ρ parameter allows for correlation between successive accident years:

- $\rho > 0$ indicates a positive correlation between successive accident years, which thickens the tails

$$\mu_{w,d} = \log(\text{Premium}_w) + \text{logelr} + \alpha_w + \beta_d + \rho(\log(C_{w-1,d}) - \mu_{w-1,d})$$

- **Incurred CAY** – Corrects the CRC model's light tail and better estimates ultimate losses

Integrated CSR Paid & CAY Incurred Models (IPI)

Key improvement over CSR and CAY:

The IPI model combines the CSR Paid and CAY Incurred models and fits the joint model on both the paid and incurred triangles simultaneously, using the same logelr and α_w parameters.

$$_P\mu_{w,d} = \log(\text{Premium}_w) + \text{logelr} + \alpha_w + {}_P\beta_d \times (1 - \gamma)^{w-1}$$

$$_I\mu_{w,d} = \log(\text{Premium}_w) + \text{logelr} + \alpha_w + {}_I\beta_d + \rho(\log({}_IC_{w-1,d}) - {}_I\mu_{w-1,d})$$

- **Paid IPI** – Similar predictive accuracy to CSR with lower variance
- **Incurred IPI** – Similar predictive accuracy to CAY with lower variance

The predictive accuracy is comparable to the CSR/CAY models, but this approach reduces the standard error around the logelr and α_w parameters and around the best estimate. Because the IPI model more accurately estimates ultimate loss with lower uncertainty, it reduces the risk margin (a key benefit).

Lower Estimation Variance → Less Required Capital → Smaller Cost-of-Capital Risk Margin

Risk Margin & Dependencies

Dependencies between lines of business

The paper uses fitted CSR and IPI models to analyze dependencies. The results are:

- Prospective analysis of the upper triangle ($elpd_{loo}$ statistic) supports independence between lines
- Retrospective analysis of the lower triangle ($elpd_{test}$ statistic) supports dependence between lines

This suggests there may be unknown variables causing dependency in the holdout time period. This means that when we calculate diversification credits, we should assume some amount of dependence between lines.

Reserve Liability and Cost of Capital Risk Margin

A key goal in this paper is to estimate loss reserve liability:

$$\text{Reserve Liability} = \text{Best Estimate} + \text{Risk Margin}$$

Best estimate (E_{Best}) - The discounted expected future loss payments (unpaid loss) as of $t = 0$, averaged over all simulations from the model, and using the risk-free rate (i).

Capital - The amount held above the best estimate to cover uncertainty and risk of adverse development. Capital grows at the risk-free rate (i)

Risk margin - The discounted cost of capital to investors as losses are paid out and capital is released, using the risky rate (r) for discounting.

$$R_{COC} = K_0 - \sum_{t=1}^{u+1} \frac{K_{t-1} \times (1+i) - K_t}{(1+r)^t} \quad \text{Risk Margin} = \text{Capital}_{\text{Initial}} - \sum_{CY} \frac{\text{Released Capital}_t}{(1 + \text{risky rate})^{CY}}$$

Model Conclusions

The tables below show the retrospective analysis results of the reputation for each model:

<u>Paid Loss Models</u>		<u>Incurred Loss Models</u>	
Model	Result	Model	Result
Mack	Biased High	Mack	Light-tailed
Bootstrap ODP	Biased High	CRC	Light-tailed
CRC	Biased High	SCC	Light-tailed (worst)
SCC	Light-tailed (worst)	CAY	Validates
CSR	Validates	IPI	Validates (lower std error)
IPI	Validates (lower std error)		

Suggested Process for a Loss Reserve Analysis

1. Fit the Mack model on paid and incurred loss data, and review triangle data reasonableness
2. Fit CRC & CSR on paid data and CRC & CAY on incurred data with wide priors
3. Build standardized residual box plots and calculate the elpd_{loo} statistic for each model
4. Select a paid model and an incurred model from those diagnostics
5. Consider adjusting the prior distributions and/or modifying the model, for example:
 - a. Tighten prior distributions of the $\log \text{elr}$ and α_w parameters
 - b. Force the final β_d parameters toward zero
 - c. Allow the ρ parameter to move toward zero as d increases in the CAY model
 - d. Allow the γ parameter to change linearly as w increases in the CSR model
6. Redo the box plots, recalculate the elpd_{loo} statistics, and review them
7. Combine the final paid and incurred models in an IPI model to get a more accurate (lower-variance) estimate of ultimate liability

Recipes for Calculation Problems

- p-p Plots & Histogram
- Interpret p-p Plots
- Kolmogorov – Smirnov (K-S) Test
- Cross Classified Model (CRC)
- Stochastic Cape Cod Model (SCC)
- Changing Settlement Rate Model (CSR)
- Correlated Accident Year Model (CAY)
- Integrated Paid and Incurred (IPI) Model
- Predicted Percentile of Outcome
- Diagnostics of Model Fit
- Discounted Best Estimate of Unpaid Liability
- Cost of Capital Risk Margin
- Diversification Credit

Overview

The goal of this paper is to connect the Chain Ladder method, a deterministic model that gives a point estimate, to a set of stochastic models. It demonstrates the following:

- The Chain ladder model estimates are maximum likelihood estimates for two different stochastic models, the ODP Mack Model and the ODP Cross-Classified Model
- Those stochastic models can be represented as GLMs and used to create a predictive distribution of outstanding losses and estimate prediction error using the Delta Method and the Bootstrap

Stochastic Models (GLMs)

A GLM is specified with the following to estimate the β parameters using MLE:

- **Error distribution** – Exponential Dispersion Family (EDF)
- **Index p** – Relationship between the mean and variance
- **Explanatory variables (x_i)** – i.e., covariates or predictors (continuous or categorical)
- **Link function $h()$** , e.g., identity link, log-link

Tweedie and ODP Distributions (EDF Sub-Families)

Generalized Tweedie

$$E[Y] = \mu = [(1-p)\theta]^{\frac{1}{1-p}} \quad \text{Var}(Y) = \phi\mu^p$$

Over-Dispersed Poisson ($p = 1$)

$$E[Y] = \mu = e^\theta \quad \text{Var}(Y) = \phi\mu$$

Distribution	p	μ
Normal	0	θ
ODP	1	e^θ
Gamma	2	$-1/\theta$
Inverse Gaussian	3	$(-2\theta)^{-\frac{1}{2}}$

Residuals for Model Validation

Check different residual plots against each variable to validate the model assumptions:

Standardized Pearson residuals

$$R_i^P = \frac{Y_i - \hat{Y}_i}{\hat{\sigma}_i}$$

Standardized Deviance residuals

$$R_i^D = \text{sgn}(Y_i - \hat{Y}_i)(d_i/\hat{\phi})^{0.5}$$

- Should be random around zero (**unbiased**) and have uniform dispersion (**homoscedasticity**)
- Should be normally distributed, so they're more useful for model assessment.

Chain Ladder Model

The basic chain ladder method is the foundational model that the rest of the paper is built on.

Stochastic Models Supporting the Chain Ladder

Key Idea

The chain ladder estimate is also the **maximum-likelihood forecast** of outstanding claims for two different formal stochastic models (below). These stochastic models can be reformulated as GLMs and then used to create a predictive distribution of outstanding claims.

Non-Parametric Mack Model (Mack Chain Ladder paper)

The traditional Mack model has the following assumptions, slightly reordered from the Mack paper:

- Losses are independent between accident years (Mack assumption 2)
- For each accident year, losses form a Markov chain, meaning that the loss, X_{kj} , is only dependent on the prior development period loss, $X_{k,j-1}$ (Part of Mack Assumption 1)
- For each accident year, the next development period's losses have the following:
 - Expected loss is proportional to losses-to-date (Mack Assumption 1)
 - $E[X_{k,j+1}|X_{kj}] = f_j X_{kj}$ where f_j is an LDF
 - Variance is proportional to losses-to-date with a proportionality constant that varies by age (Mack Assumption 3)
 - $Var[X_{k,j+1}|X_{kj}] = \sigma_j^2 X_{kj}$

The Mack model is **stochastic** (because it considers the mean and variance of observations) but **non-parametric** (it does not specify a distribution of loss observations).

Key Results (Non-Parametric Mack model)

1. The conventional chain ladder LDF estimators are **unbiased** and **minimum variance** among unbiased estimators that are *linear combinations of age-to-age factors*
2. The conventional chain-ladder reserve estimator, $\hat{R}_k$, is **unbiased**.

Note:

These results are shown in Mack Chain Ladder Appendices B and C.

Parametric Mack Model

The parametric EDF Mack model has the same structure but makes one assumption change:

- It assigns an EDF distribution to the incremental losses Y_{kj} . This replaces the variance assumption in the traditional Mack model.

The **ODP Mack model** assigns an *ODP distribution* to incremental losses and is the most important sub-family of the EDF Mack model.

Theorem 3.1 (EDF Mack Model)

If the original Mack variance assumption holds, then the weighted-average chain ladder LDFs, $\hat{f}_j$, are **unbiased** estimators and MLEs (Maximum Likelihood Estimates) of the f_j parameters.

Theorem 3.1 (ODP Mack Model)

For the ODP Mack model, with column-only dispersion parameters ($\phi_{k,j} = \phi_j$), the chain ladder LDFs, $\hat{f}_j$, are *both* MLEs *and* MVUEs (minimum variance unbiased estimators) of the f_j parameters.

The cumulative loss estimates, $\hat{X}_{k,j}$, and reserve estimates, $\hat{R}_k$, are also MVUEs.

Key Takeaways from Theorem 3.1

For the ODP Mack Model, the conventional chain ladder factors and reserve estimates are minimum variance over *all unbiased estimators*, not just linear combinations of age-to-age factors. This is a stronger result than for the Non-Parametric Mack model.

The exciting takeaway is that the normal chain ladder estimates and forecasts, developed decades ago, are statistically **optimal estimators** (both MLE and MVUE) when the assumptions hold.

Cross-Classified Models

The EDF cross-classified model (also in the Meyers paper) is structured with explicit row (α_k) and column (β_j) parameters, and has the following assumptions:

- Incremental losses Y_{kj} are stochastically independent
- For each accident year and development period:
 - $Y_{kj} \sim EDF(\theta_{kj}, \phi_{kj}, a, b, c)$
 - $E[Y_{kj}] = \alpha_k \beta_j$, for $\alpha_k, \beta_j > 0$
 - $\sum_{j=1}^J \beta_j = 1$

The ODP cross-classified model also has the following assumptions:

- Y_{kj} is restricted to an ODP distribution
 - The dispersion parameters ϕ_{kj} are identical for all cells (i.e., $\phi_{kj} = \phi$)
-

Theorem 3.2 (ODP Cross-Classified Model)

If the ODP cross-classified model assumptions hold, then the MLE fitted values and forecasts $\hat{Y}_{kj}$ are the same as those from the conventional chain-ladder method.

Theorem 3.3 (ODP Cross-Classified Model)

If those assumptions hold and the fitted values $\hat{Y}_{kj}$ and reserve estimates $\hat{R}_k$ are *also corrected for bias*, then they are MVUEs of Y_{kj} and R_k .

Key Takeaways from Theorems 3.2 and 3.3

- Forecasts from the ODP Mack and ODP Cross-Classified models are *identical to those from the chain ladder method*, despite the different formulations
- We can calculate forecasts for the ODP Cross-Classified model without directly using it, simply by working as if it were an ODP Mack model

GLM Representation of the Models

The ODP Mack and ODP Cross-Classified models can be represented as GLMs, and the estimates and forecasts from their fitted parameters will match those from the chain ladder method. The benefit is that these GLMs can be used to analyze prediction errors and create a predictive distribution of reserves.

	<u>ODP Mack Model</u>	<u>ODP Cross-Classified Model</u>
Structure	$Y_{k,j+1} X_{kj} \sim ODP((f_j - 1)X_{kj}, \phi_{kj})$	$Y_{kj} \sim ODP(\alpha_k \beta_j, \phi)$ and $\phi_{kj} = \phi$
GLM Form	$\hat{f}_{kj} - 1 X_{kj} \sim ODP(f_j - 1, \frac{\phi_j}{X_{kj}})$ • Identity link function • $w_{k,j} = X_{k,j}$ weights GLM gives estimates of $f_j - 1$	$Y_{kj} \sim ODP(\mu_{kj}, \phi)$ $\mu_{kj} = e^{\ln(\alpha_k) + \ln(\beta_j)}$ • Log link function • $w_{k,j} = 1$ weights GLM gives estimates of α_k and β_j

Prediction Error

Forecast (prediction) error splits into three components:

- **Parameter error** - The error due to the uncertainty in the fitted parameters
- **Process error** - The irreducible noise in future claims due to the insurance process
- **Model error** - The error due to the difference between the model specification and the correct model (not separately estimable, so it's absorbed into process error)

Mean Square Error of Prediction (MSEP)

MSEP is a measure of forecast error. When parameter and process error are independent:

$$MSEP[e_{(r)}^*] = E \left\{ [e_{(r)param}^*]^2 \right\} + E \left\{ [e_{(r)process}^*]^2 \right\}$$

$$MSEP = E\{[parameter\ error]^2\} + E\{[process\ error]^2\}$$

Goodness of Fit and Model Complexity

Improving a model's goodness-of-fit doesn't necessarily decrease its MSEP. If a model has too many parameters, it can overfit the data and result in a worse MSEP when applied to *test data* (holdout data).

There are a few proxy measures of prediction error used to penalize overfitting when comparing models:

- **Information Criteria (AIC, BIC)** – Model fit error on the training data penalized for # parameters
- **Generalized Cross-Validation** – An approximation to leave-one-out cross validation

Prediction Error of Outstanding Losses (Delta Method and The Bootstrap)

The delta method and bootstrap take the ODP Mack model, and particularly the ODP Cross-Classified model, and use them to estimate prediction error (parameter error + process error) for outstanding losses.

Delta Method

The Delta Method uses the parameter estimates and standard errors from the ODP Cross-Classified model (the α_k and β_j parameters) along with the covariance matrix of those GLM parameter estimates to estimate the variance of the total outstanding losses, $\hat{R}$. But, it gives no distribution, so a distribution (e.g. Lognormal) must be assumed and fit to the mean and variance.

Advantages

- Computationally simple
- Produces an estimate of MSEP

Disadvantages

- Unknown accuracy
- Gives no distribution of the prediction error
- Only gives a second order approximation

Note:

The total reserve MSEP exceeds the sum of the individual accident year MSEPs, because every accident year uses the same development parameters, so they are positively correlated. This is also pointed out in the Mack Chain Ladder paper.

Bootstrap

The Bootstrap method is computationally more demanding, but it produces the *entire predictive distribution* of the estimated outstanding losses by resampling residuals with replacement to create many pseudo-triangles. The model is refit to each pseudo triangle to create a pseudo forecast, and process variance is added.

Taylor's Semi-Parametric GLM bootstrap is the same as the Shapland bootstrap, except for how process error is handled:

- Shapland samples from a gamma distribution for future losses
- Taylor uses a second residual resampling

The parametric bootstrap is similar, but is based on theoretical distributions. The parametric bootstrap is also discussed in Shapland.

Numerical Examples

Make sure to review and understand the numerical examples in the study guide (and paper).

Model Validation

Three Aspects of Model Validation

- Analysis of **distributional assumptions**
 - Testing whether the model's distributional assumptions appropriately represent the data
- Analysis of **goodness-of-fit**
 - Testing how well the model fits to the data and to make sure we include all significant drivers of the target variable
- Analysis of the model's **predictive performance on new data**
 - We can't analyze that from a triangle, but Meyers evaluates this with retrospective analysis

Diagnostic Graphs

Diagnostic graphs are analyzed based on *deviance residuals*, not Pearson residuals.

- **Scatterplot of Residuals** – Plots residuals against a relevant variable (expected value, accident year, development period, or calendar year)
 - Plots should be random around zero and homoscedastic
 - A trend suggests goodness-of-fit issues
 - Heteroscedasticity suggests the dispersion assumption is inappropriate
- **Spread Plot** – Plots the inter-quartile range of residuals and the standard deviation of residuals
 - Widening or narrowing of the inter-quartile range suggests heteroscedasticity
 - Std. Dev should fluctuate around 1
- **Actual & Expected Comparison Plot** – Shows actual and expected totals by AY, development period, and calendar year
 - Actual and expected values should be close → Significant differences suggest poor model fit
- **Actual & Expected Ratio Plot** – Actual/expected ratio plot vs. AY, dev period, and calendar year
 - Ratios should be random around 100% → Significant deviations/trends suggest poor model fit
- **Actual & Expected Ratio 2-D Heat Map** – Actual/expected ratio by cell
 - Colors should be random → Significant clusters indicate poor fit and possible missing terms
- **P-P plot** – Plots the theoretical, selected distribution vs the empirical distribution
 - Should follow a 45° line → Significant deviations suggest issues with distributional assumptions
- **Residual histogram** – Standardized deviance residuals should be approximately standard normal

Analysis of Goodness-of-Fit

First, we use goodness-of-fit tests to build a model that fits well. Analyze the following diagnostic graphs by accident year, development period, and calendar year to test for unmodeled trends:

- Scatterplot of deviance residuals

- Actual & expected comparison plot
- Actual & expected ratio plot

Analyze the heat map for clusters that suggest missing interactions or calendar year effects.

Note:

The ODP fit forces the actual and expected totals to match along the fitted dimensions in the model (i.e. the rows and columns). Because of this, an actual vs expected plots by a fitted dimension (accident year and development period) will be in complete agreement, so they provide no information. Instead, we need to analyze the residuals for non-fitted dimensions like calendar year.

Analysis of Distribution Assumptions

After model fitting and checking goodness-of-fit, we need to test the underlying distributional assumptions and make any necessary adjustments. Below are the key assumptions to test:

- The form of the distribution of the data
- The scale parameter of the distribution
- The selected link function

Analyze the following diagnostic graphs to test distributional assumptions:

- Residual plots (scatterplot, spread plot)
- p-p plot

Heteroscedastic residuals suggest that the scale parameter assumption is incorrect and should be adjusted using weights or that a different distribution should be considered.

There's no formal test for the link function, so it's determined by model structure:

- Log link → multiplicative model
- Identity link → additive

Recipes for Calculation Problems

- ODP Cross-Classified Model Parameters
- GLM Representation of Chain Ladder Models
- Re-Normalizing ODP Cross-Classified Parameters
- Delta Method for Reserve Prediction Error
- Semi-Parametric Bootstrap Method

Predictive Distributions for Reserves which Incorporate Expert Opinion

Overview

In traditional loss reserving with deterministic methods, it's common to select LDFs or select an expected loss ratio for the BF method. These are ways to incorporate expert opinion. However, if we want to calculate a reserve range using a traditional bootstrap model or Mack model, making LDF selections may compromise some of the model assumptions.

A key benefit of using a Bayesian approach (compared to a Mack or bootstrap model) is that we can naturally incorporate expert opinion within a stochastic model and get a reserve range without compromising the underlying assumptions.

Examples of when to incorporate expert knowledge or opinion

- Change in payment pattern due to a change in company policy
- Change in benefits due to new laws, requiring adjustment to LDFs

Areas where expert knowledge might be used

- Selecting LDFs for the chain ladder method
 - Overriding LDFs in a particular row (accident year)
 - Using only n-years to estimate the LDFs
- Selecting expected loss for the BF method

Different Stochastic Models for Chain Ladder

Each of the models below give the same reserve result as the chain ladder method, but they differ in form and may have different variances.

Mack Model

$$E[D_{i,j}] = \lambda_j \cdot D_{i,j-1}$$

$$\text{Var}(D_{i,j}) = \sigma_j^2 \cdot D_{i,j-1}$$

$$E[Loss_{AY,k}] = LDF_k \cdot Loss_{AY,k-1}$$

$$\text{Var}(Loss_{AY,k}) = \sigma_j^2 \cdot Loss_{AY,k-1}$$

Advantages:

- Easy to implement – Parameter estimates and prediction errors can be calculated in a spreadsheet.

Disadvantages:

- There is no predictive distribution, because a distribution isn't specified.

- Need to estimate separate parameters, σ_j^2 , for the variance apart from the LDFs.

Over-Dispersed Poisson Models

GLM approach (similar to Shapland GLM Bootstrap):

$$\mathbb{E}[C_{i,j}] = m_{i,j} = e^{c+\alpha_i+\beta_j}$$

$$\mathbb{E}[IncLoss_{AY,k}] = e^{c+\alpha_i+\beta_j}$$

Row-Column Form:

$$\mathbb{E}[C_{i,j}] = x_i \cdot y_j \quad \text{and} \quad \sum y_j = 1$$

$$\mathbb{E}[IncLoss_{AY,k}] = Row\ Factor_{AY} \cdot Col\ Factor_k$$

x_i – Expected ultimate loss for accident year i up to the last development period of the triangle
 y_j – % of ultimate loss emerging in development period j

Advantages:

- Doesn't necessarily break down if there are some negative incremental values.
- It gives the same reserve estimate as the chain ladder method.
- It's more stable than the log-normal model of Kremer.

Disadvantages:

- Connection to the chain ladder method is not immediately apparent.

Over-Dispersed Negative Binomial

This model differs in form from the ODP model, but produces the same predictive distribution.

$$\mathbb{E}[C_{i,j}] = (\lambda_j - 1) \cdot D_{i,j-1}$$

$$\mathbb{E}[IncLoss_{AY,k}] = (LDF_k - 1) \cdot Loss_{AY,k-1}$$

$$\text{Var}(C_{i,j}) = \phi \cdot \lambda_j \cdot (\lambda_j - 1) \cdot D_{i,j-1}$$

$$\text{Var}(IncLoss_{AY,k}) = dispersion \cdot LDF_k \cdot (LDF_k - 1) \cdot Loss_{AY,k-1}$$

Advantages:

- Doesn't necessarily break down if there are some negative incremental values (but the sum of the column needs to be positive).
- It gives the same reserve estimate and has the same form as the chain ladder method.

Disadvantages:

- Column sums of incremental losses must be positive (or variance would be negative).

Normal Distribution

$$\mathbb{E}[C_{i,j}] = (\lambda_j - 1) \cdot D_{i,j-1}$$

$$\mathbb{E}[IncLoss_{AY,k}] = (LDF_k - 1) \cdot Loss_{AY,k-1}$$

$$\text{Var}(C_{i,j}) = \phi_j \cdot D_{i,j-1}$$

$$\text{Var}(IncLoss_{AY,k}) = dispersion_k \cdot Loss_{AY,k-1}$$

Advantages:

- Can handle negative incremental losses.

Disadvantages:

- Need to estimate separate parameters, ϕ_j , for the variance apart from the LDFs.

Prediction Error

Given loss data, we can use one of the models above to create a distribution of future losses. The expected value of the distribution is our prediction, the central estimate. The root mean square error (RMSEP) is the prediction error, incorporating both process variance and parameter (estimation) variance.

$$\boxed{\text{prediction variance} = \text{process variance} + \text{estimation variance}}$$

Prediction Error vs. Standard Error

Prediction error accounts for uncertainty in the parameter estimation (estimation variance) and the variability of future losses (process variance). Standard error only accounts for estimation variance.

$$\text{prediction error} = \sqrt{\text{process variance} + \text{estimation variance}} \qquad \text{standard error} = \sqrt{\text{estimation variance}}$$

Incorporating Expert Opinion in Chain Ladder

The over-dispersed negative binomial model can allow for a separate LDF parameter for each accident year and development period. This flexibility allows us to incorporate expert opinion in selecting LDFs.

Incorporating Expert Opinion to Override Specific LDFs

- For each LDF parameter that is selected, specify the prior distribution of $\lambda_{i,j}$:

$$E[\lambda_{i,j}] = LDF^{\text{selection}} \qquad \text{Var}(\lambda_{i,j}) - \text{set to be relatively small}$$

- Group the other LDF parameters by development period to reflect the chain ladder method, by setting $\lambda_{i,j} = \lambda_j$ and using a large variance for the prior distribution.

Incorporating Expert Opinion to Use n -year LDFs

$$\lambda_{i,j} = \lambda_j \qquad \text{for the most recent } n \text{ calendar year diagonals}$$

$$\lambda_{i,j} = \lambda_j^* \qquad \text{for all diagonals prior to the latest } n \text{ diagonals}$$

Set the prior distributions of the LDF estimates λ_j and λ_j^* with large variances so that they're estimated from the data.

How Variance of the Prior Distribution Impacts the Model Results

- “Vague” priors (large variance) – Result is close to what the data estimates (chain ladder method)
- “Strong” priors (small variance) – Result is drawn toward the mean of the prior distribution

Bayesian Model for Bornhuetter-Ferguson

The Bayesian model of the BF method is based on the over-dispersed Poisson distribution (row-col form). For the BF model, select the expected ultimate loss by accident year by setting a prior distribution for each of the row parameters, x_i :

$$x_i \sim \text{Gamma}(\alpha_i, \beta_i)$$

$$\boxed{E[x_i] = \frac{\alpha_i}{\beta_i} = M_i}$$

$$\boxed{\text{Var}(x_i) = \frac{\alpha_i}{\beta_i^2} = \frac{M_i}{\beta_i}}$$

Credibility-Weighted Bayesian Model for the BF Method

The Bayesian model is a credibility model between the chain ladder and Bornhuetter-Ferguson methods. We can adjust the credibility by changing β_i :

- Increasing β_i causes the variance of the prior distribution to decrease (indicating higher confidence in the selected ultimate loss).
- This causes $Z_{i,j}$ to decrease placing more weight on the BF method.

$$E[C_{i,j}^{CL}] = (\lambda_j - 1) \cdot D_{i,j-1}$$

$$E[C_{i,j}^{BF}] = M_i \cdot \%IncPaid_j$$

$$Z_{i,j} = \frac{p_{j-1}}{\beta_i \cdot \phi + p_{j-1}}$$

$$E[C_{i,j}^{cred}] = Z_{i,j} \cdot E[C_{i,j}^{CL}] + (1 - Z_{i,j}) \cdot E[C_{i,j}^{BF}]$$

Fully Stochastic Bayesian Model for the BF Method

The model above is stochastic in the row parameters, x_i , but uses the fixed LDFs directly.

The fully stochastic model uses an improper gamma prior distribution for the column parameters and the prior distribution shown above for the row parameters, x_i . The goal here is to have a model that is fully stochastic in both row and column parameters.

How Variance of the Prior Distribution Impacts the Model Results

- “Vague” priors for x_i (large variance) – Result is close to the chain ladder method
- “Strong” priors for x_i (small variance) – Result is close to the Bornhuetter-Ferguson method

The flexibility of the fully stochastic model means we can replicate the chain ladder method, the BF method or a weighting between the two methods by adjusting the variance of the prior distributions of the row parameters.

Advantage of a Stochastic Approach

- We can produce the full predictive distribution of the reserve, not just the point estimate or the mean and standard deviation.

Recipes for Calculation Problems

- Incorporating Expert Opinion: Chain Ladder
- Bayesian Model for the BF Method
- Fully Stochastic BF Model
- Estimated Reserve from Fully Stochastic BF

Overview

The main goal of the Marshall paper is to answer the question of how to set an appropriate risk margin for insurance liabilities. For this paper, make sure to focus on the sources of uncertainty (independent risk and internal/external systemic risk). Also, make sure you thoroughly understand the risk margin calculation and the different types of additional analyses.

Preparing the Claims Portfolio

Splitting the Claims Portfolio to Valuation Classes

It's preferable to split the claims portfolio into the same valuation classes that are used in the reserve analysis to determine central estimates.

Splitting Valuations Classes to Claims Groups

There should be homogeneity within a claims group and different behavior between claims groups (e.g. split the Home portfolio between CAT and non-CAT claims).

Sources of Uncertainty

Independent Risk – Risks arising from randomness inherent in the insurance process

Systemic Risk – Risks that are potentially common across valuation classes or claims groups

- **Internal Systemic Risk** – Uncertainty because the valuation models are imperfect representations of the insurance process. These are risks internal to the liability valuation process.
- **External Systemic Risk** – Risks external to the liability valuation process

Sources of Independent Risk

- Random component of *parameter risk* – Uncertainty in selecting appropriate parameters due to the randomness of insurance
- Random component of *process risk* – Pure effect of randomness due to the insurance process

Sources of Internal Systemic Risk

Specification Error – Error from the inability to make a model that fully represents the insurance process

Parameter Selection Error – Error that we can't adequately measure all the predictors (parameters) of claim cost outcomes and their trends

Data Error – Error due to poor or unavailable data that's needed for the reserve analysis

Sources of External Systemic Risk

- **Economic and Social Risks** – Normal inflation and social/environmental trends

- **Legislative, Political and Claims Inflation Risk** – Changes to the legislative or political environment affecting claims valuation portfolios and changes to claims settlement levels
- **Claim Management Process Change Risk** – Changes in the claims department that impact reporting/payment patterns or the estimating of case reserves
- **Expense Risk** – Uncertainty in expenses for managing the liabilities
- **Event Risk (Catastrophes)** – Uncertainty from losses due to natural or man-made catastrophes
- **Latent Claim Risk** – Uncertainty from losses that may arise in the future from a source that's not considered covered (e.g. asbestos)
- **Recovery Risk** – Uncertainty in reinsurance (or non-reinsurance) recoveries

Quantitative vs. Qualitative Analysis

- Quantitative analysis can only reflect uncertainty in historical experience and can't adequately reflect all possible sources of future uncertainty.
- We also need qualitative analysis (judgment) to make sure we reflect all sources of uncertainty.

Quantitative modeling works best to analyze sources of independent risk and past episodes of external systemic risk. Quantitative modeling needs to be supplemented with other quantitative and qualitative analyses to reflect internal systemic risk and future external systemic risk that differs from the past.

Correlation Effects

Correlation for Independent Risks

Independent risks are uncorrelated with other sources of uncertainty both within and between valuation classes.

Correlation for Internal Systemic Risks

- Uncorrelated with independent risk and external systemic risk sources
- Correlated between classes or between outstanding claim and premium liabilities due to:
 - **Same actuary effect** – Effect of common valuation approaches for different valuation classes
 - Link between premium liability methodology and outcomes from outstanding claim valuation

Correlation for External Systemic Risks

- Uncorrelated with independent, internal systemic risk and generally uncorrelated with other external systemic risk categories (e.g. event risk is uncorrelated with economic/social risk)
- Correlated between classes or between outstanding claim and premium liabilities in the same risk category
- "It is possible that external systemic risk categories may be partially correlated either within or between valuation classes"

Quantitative Methods for Assessing Correlation

- Quantitative techniques are complex and require lots of data – time and effort outweigh benefits
- Correlations would be driven by past correlations, but correlations for external systemic risk sources may be different than past episodes.
- It's difficult to separate past correlations between independent and systemic risks.
- Internal systemic risk can't be modeled with standard correlation techniques.
- Results are unlikely to be aligned with the framework (split between independent and internal/external systemic risk)

Risk Margin Calculation

To calculate the risk margin, we calculate the CoVs separately for independent risk, internal systemic risk, and external systemic risk. Then, we consolidate the CoVs assuming independence between the sources of risk and calculate the risk margin (assuming the normal or lognormal distribution).

Additional Analysis

Sensitivity Testing

Goal:

Gain insights about the sensitivity of the final risk margin to key assumptions.

Do this by varying each key assumption (the CoVs and correlations) and monitor the impact on the risk margin. Review key assumptions that have significant impact on the risk margin.

Scenario Testing

Goal:

Tie the risk margin to a set of valuation outcomes. We adjust the assumptions used for the central estimate in order to align the central estimate with the assessed provisions including the risk margin.

Scenario testing asks the question: What scenario (e.g. higher frequency or severity) would result in the central estimate increasing to the current central estimate plus the risk margin?

Internal Benchmarking

Goal:

Compare the proposed CoVs to one another to check for internal consistency and reasonableness.

Independent Risk CoV Selection:

- **Portfolio size:** Larger portfolios have lower CoVs due to lower volatility from random effects
- **Length of Claim Runoff:** Longer tailed lines have higher CoVs due to more time for random effects to have an impact

Implications:

- **Long-tailed portfolios:** Premium Liability CoV should be higher than Outstanding Claim CoV

- **Short-tailed portfolios:** Premium Liability CoV should be lower than Outstanding Claim CoV

Internal Systemic Risk CoV Selection:

- If template models are used for similar valuation classes, we would expect similar CoVs.
- If similar valuation methodology is used on both short and long-tail classes, then we would expect a higher CoV for the long-tail class.

External Systemic Risk CoV Selection:

- CoVs should be higher for long-tail portfolios, except for event risk for Property and liability risk for Home

External Benchmarking

Goal:

Compare the reasonableness of CoVs and risk margins to those from external sources.

Shouldn't be relied on instead of analysis, but may be useful when there is little information available for analysis, especially for independent risk.

Hindsight Analysis

Goal:

Compare the past reserve estimates of liabilities against the latest view of the same liabilities to review the actual volatility in the past.

Mechanical Hindsight Analysis:

1. Apply the chain ladder method to data as of the valuation date to calculate a current unpaid estimate.
2. Iteratively, remove a diagonal of data and apply the same method to calculate unpaid estimates for prior valuation dates.
3. Compare the past estimates to the current estimate for the same accident years. The relevant payments made between the past valuation dates and the current valuation date should be added to the current unpaid estimate.

Independent Risk Assessment

The goal of the independent risk assessment is to use a model to “fit away” past systemic episodes in order to analyze the residual volatility in order to get the CoVs for independent risk.

Methods to Analyze Independent Risk

- Mack method (Mack 1994 paper)
- Bootstrapping (Shapland paper)
- Stochastic Chain Ladder
- GLM techniques

- Bayesian techniques (Meyers paper)

Balanced Scorecard (Internal Systemic Risk Assessment)

Goal: Assess the adequacy of the modeling infrastructure and its ability to reflect and predict the underlying insurance process.

Potential Risk Indicators

Page 33 in the paper has a full list of the potential risk indicators. Below are a few examples:

Specification Error:

- Number of independent models used
- Range of results produced by the models
- Checks made on reasonableness of results

Parameter Selection Error:

- Best predictors have been identified
- Best predictors are stable over time (or change due to process changes)
- Value of the predictors used (predictors used are close to the best predictors)

Data Error:

- Good knowledge of past processes affecting predictors
- Extent, timeliness, consistency and reliability of information from business
- There are appropriate reconciliations and quality control for the data

Reviewing the CoV Scale

- Minimum CoV for best practice is unlikely to be less than 5%.
- Maximum CoV for worst practice could be greater than 20% (e.g. single, aggregated model with limited data)
- Scale should not be linear – Model improvements show diminishing returns
 - Improvement from a poor to fair model is greater than from a fair to good model
- CoVs for long-tail lines are higher than short-tail CoVs for the same score
 - It's more difficult to model the underlying process of a long-tail line and predictors are less stable.
- It's reasonable to use the same scale for outstanding claim liabilities and premium liabilities.

Types of External Systemic Risk

We can identify key potential sources of systemic risk through discussions with business experts as part of the valuation process. These discussions should consider:

- Underwriting and risk selection
- Pricing
- Claims Management
- Expense Management
- Emerging portfolio trends
- The environment in which the portfolio operates

CoVs for each external systemic risk category should be selected using a mix of quantitative analysis and qualitative judgment.

Economic and Social Risk

Includes normal inflation (CPI) and economic conditions (unemployment, GDP growth, ...)

Legislative, Political and Claims Inflation Risk

These risks are likely to be more significant for long-tail lines. Some examples include:

- Impact of recent legislation or changes in court interpretation
- Potential future legislation that impacts losses retrospectively
- Precedent setting in courts
- Changes in medical technology costs or legal costs
- Systemic shifts in large claim frequency or severity

Claim Management Process Change Risk

Changes can impact all valuation classes, but are more relevant for outstanding claims liabilities than premium liabilities. It's important to discuss current and future process changes with claims management and how changes could impact reporting/payment patterns, closing and reopening rates, and case reserves.

Expense Risk

Expense risk is generally a small contributor to total external systemic risk.

Event Risk (Catastrophes)

Catastrophes have the most significant impact on property lines of business.

Sources to help quantify the impact of event risk for premium liabilities:

- Past experience from event risk losses
- CAT models
- Reinsurance intermediaries

Latent Claim Risk

Latent claim risk may be material for workers compensation and liability classes, but is negligible for most valuation classes. This risk has low probability, but high potential severity (e.g. asbestos). Discussions with underwriters can help to identify sources of risk.

Recovery Risk

Likely to be insignificant for most valuation classes, except potentially for third party recoveries for Auto. Reinsurance recovery risk could be driven by catastrophes.

Recipes for Calculation Problems

- Risk Margin Calculation
- Internal Risk Balanced Scorecard